

Understanding Machine Learning: From Mathematics to Applications

Lecture 5:

Unsupervised learning and PCA

Jean Feng

Announcements

- HW2 due at midnight.
- HW3 due in 2 weeks from today. Based on lecture today.
- Think about your projects!

Unsupervised learning

- Today we'll switch gears to **unsupervised learning**.
- **Q:** What's the difference between unsupervised learning and supervised learning?
- **A:** Unsupervised learning makes fewer assumptions about the data generating mechanism than supervised learning.
- **B:** Unsupervised learning means the ML algorithm can learn without human supervision, whereas supervised learning does.
- **C:** Unsupervised learning means the ML algorithm does not have access to labelled data.

Unsupervised learning

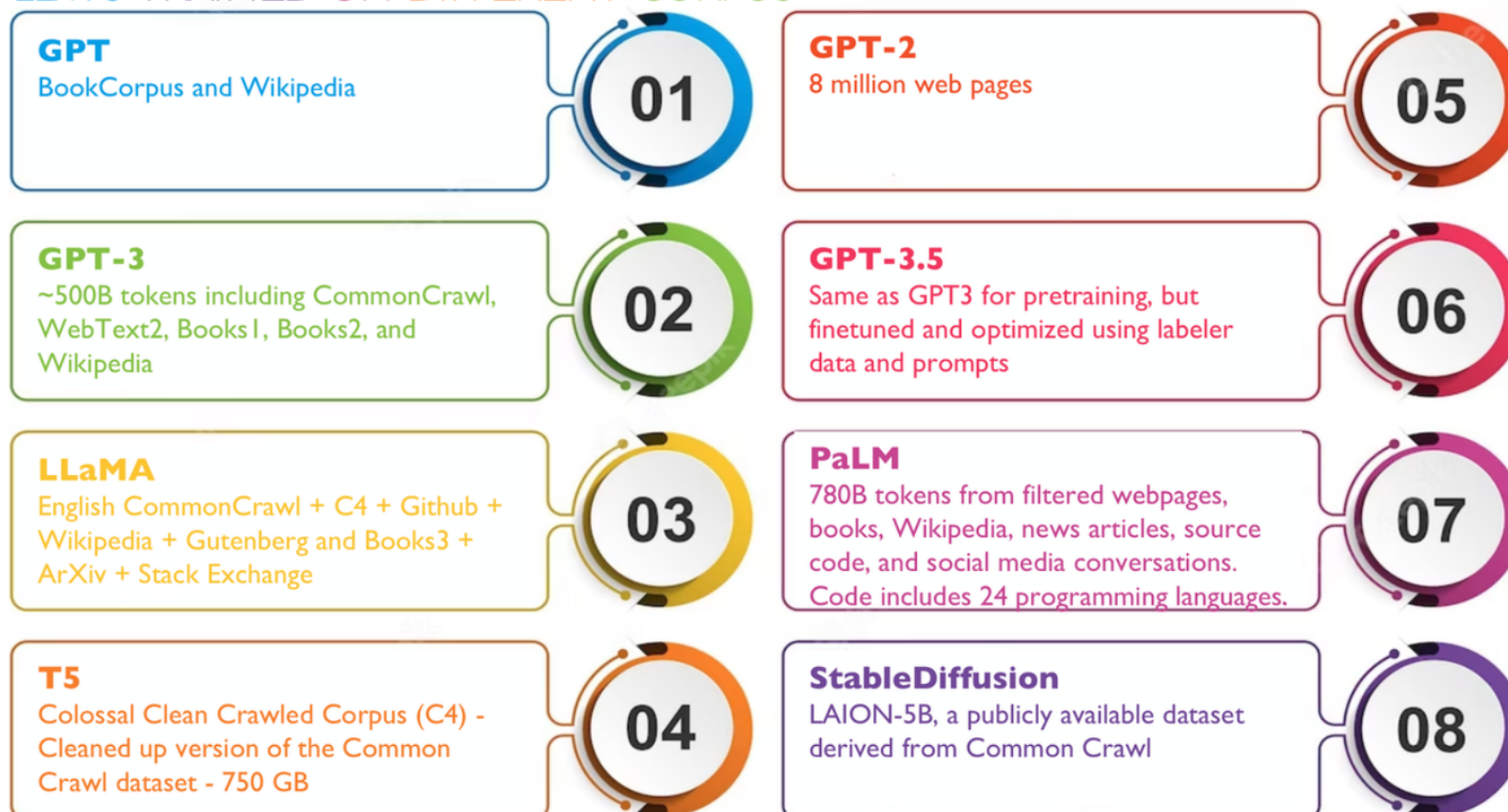
- ... is currently all the rage.



Unsupervised learning

- LLMs are typically pre-trained using “**self-supervised learning**”, which is a type of unsupervised learning.

LLM'S TRAINED ON DIFFERENT CORPUS



Unsupervised learning



Machine learning

🌐 81 languages ▾

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From Wikipedia, the free encyclopedia



For the journal, see [Machine Learning \(journal\)](#).

"Statistical learning" redirects here. For statistical learning in linguistics, see [statistical learning in language acquisition](#).

Machine learning (ML) is a [field of study](#) in [artificial intelligence](#) concerned with the development and study of [statistical algorithms](#) that can learn from [data](#) and [generalize](#) to unseen data, and thus perform [tasks](#) without explicit [instructions](#).^[1] Recently, [artificial neural networks](#) have been able to surpass many previous approaches in performance.^{[2][3]}

ML finds application in many fields, including [natural language processing](#), [computer vision](#), [speech recognition](#), [email filtering](#), [agriculture](#), and medicine.^{[4][5]} When applied to business problems, it is known under the name [predictive analytics](#). Although not all machine learning is [statistically](#) based, [computational statistics](#) is an important source of the field's methods.

The mathematical foundations of ML are provided by [mathematical optimization](#) (mathematical programming) methods. [Data mining](#) is a related (parallel) field of study, focusing on [exploratory data analysis](#) (EDA) through [unsupervised learning](#).^{[7][8]}

From a theoretical viewpoint, [probably approximately correct \(PAC\) learning](#) provides a framework for describing machine learning.

Part of a series on
**Machine learning
and data mining**

Paradigms [\[show\]](#)

Problems [\[show\]](#)

Supervised learning [\[show\]](#)
([classification](#) • [regression](#))

Clustering [\[show\]](#)

Dimensionality reduction [\[show\]](#)

Structured prediction [\[show\]](#)

Anomaly detection [\[show\]](#)

Artificial neural network [\[show\]](#)

Reinforcement learning [\[show\]](#)

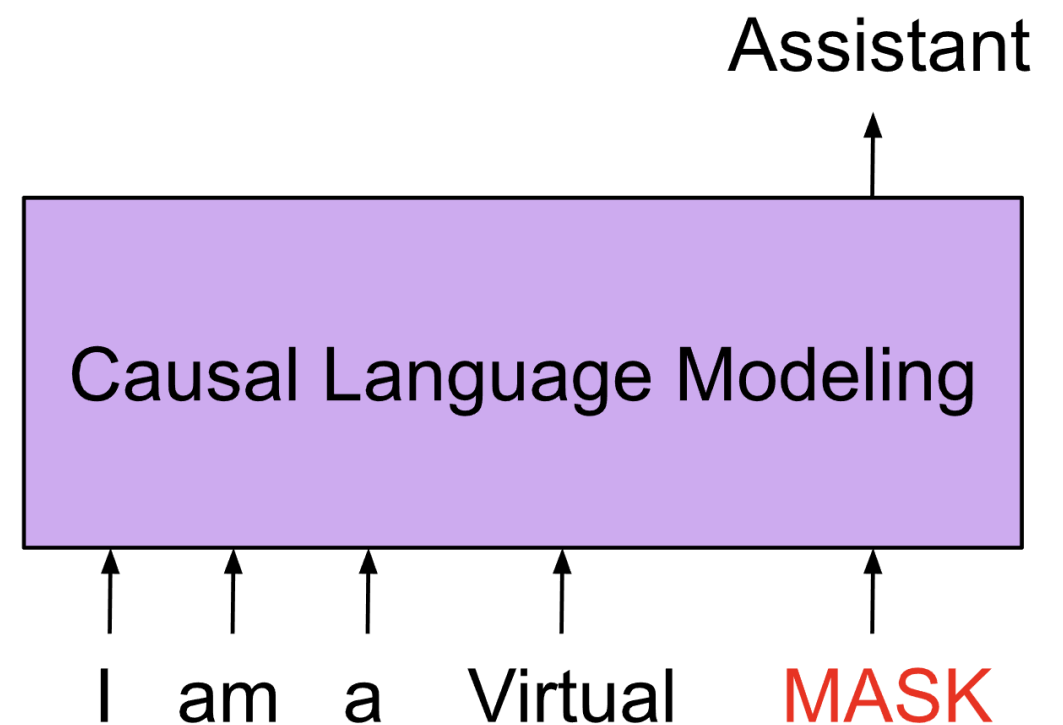
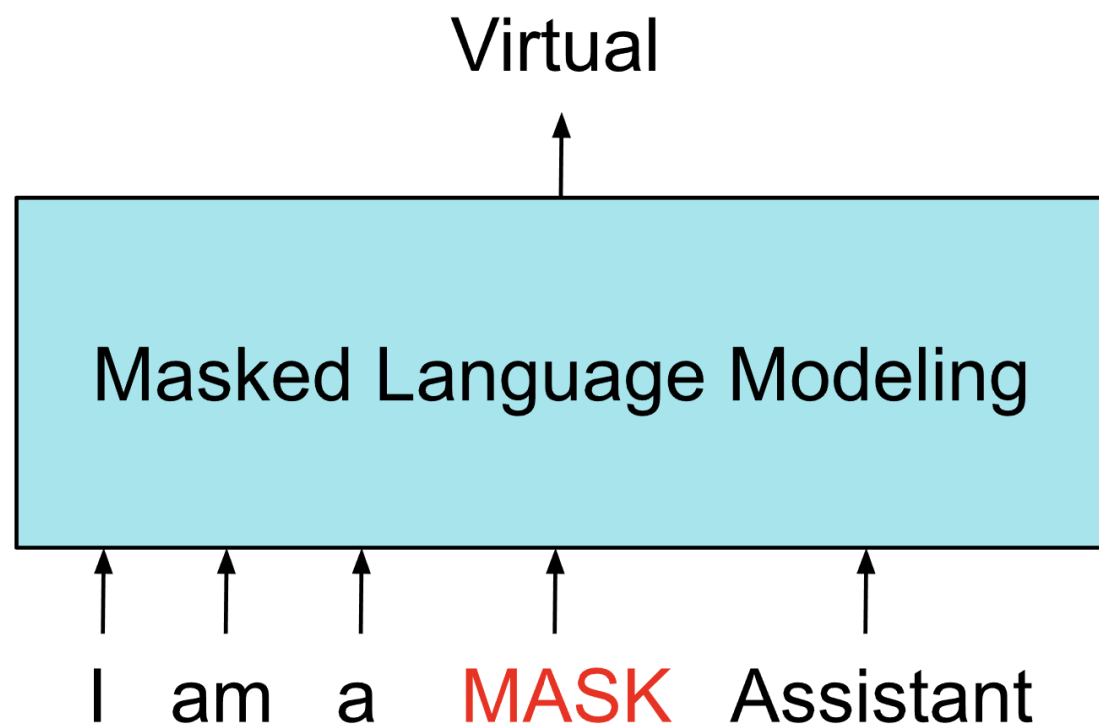
Learning with humans [\[show\]](#)

Model diagnostics [\[show\]](#)

No labels!

Unsupervised learning

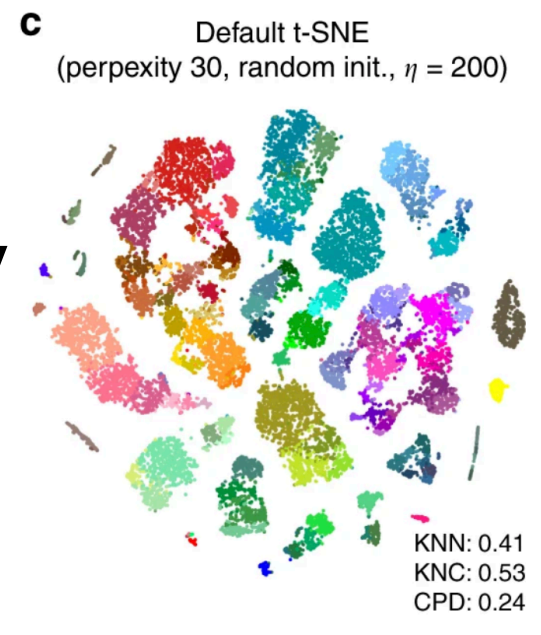
- Given that texts from the internet don't have labels, how do LLMs learn to converse like a human?
- Step 1 of many: Learn words that typically appear in the middle or at the end of a sentence.



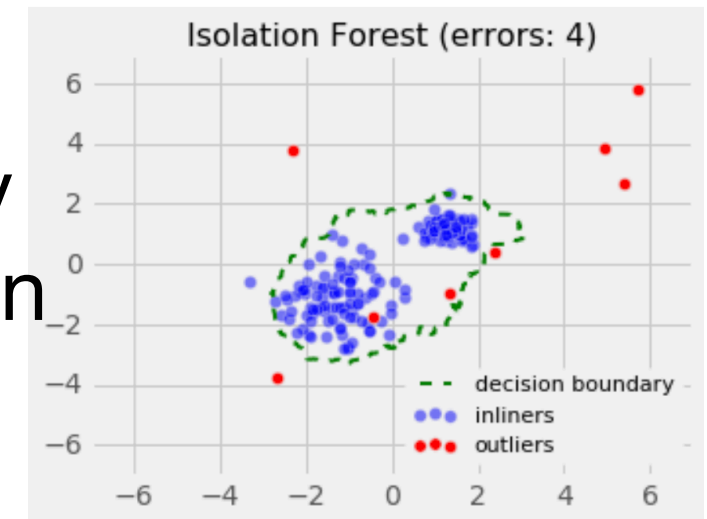
“Generative AI”

Unsupervised learning tasks

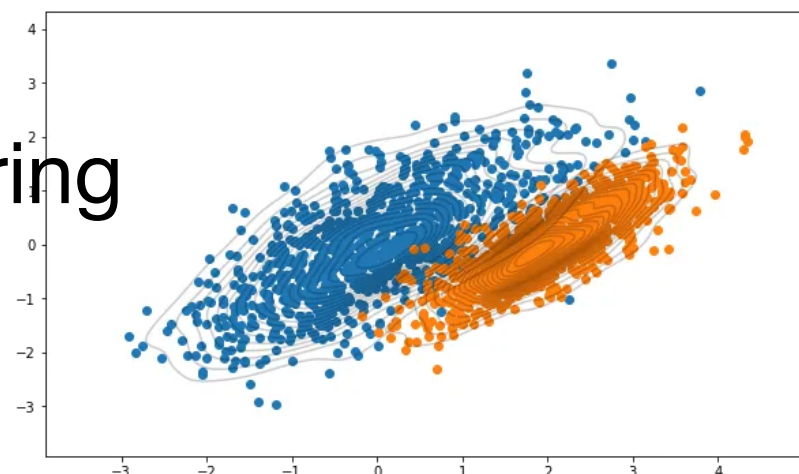
Dimensionality
Reduction



Anomaly
Detection



Clustering



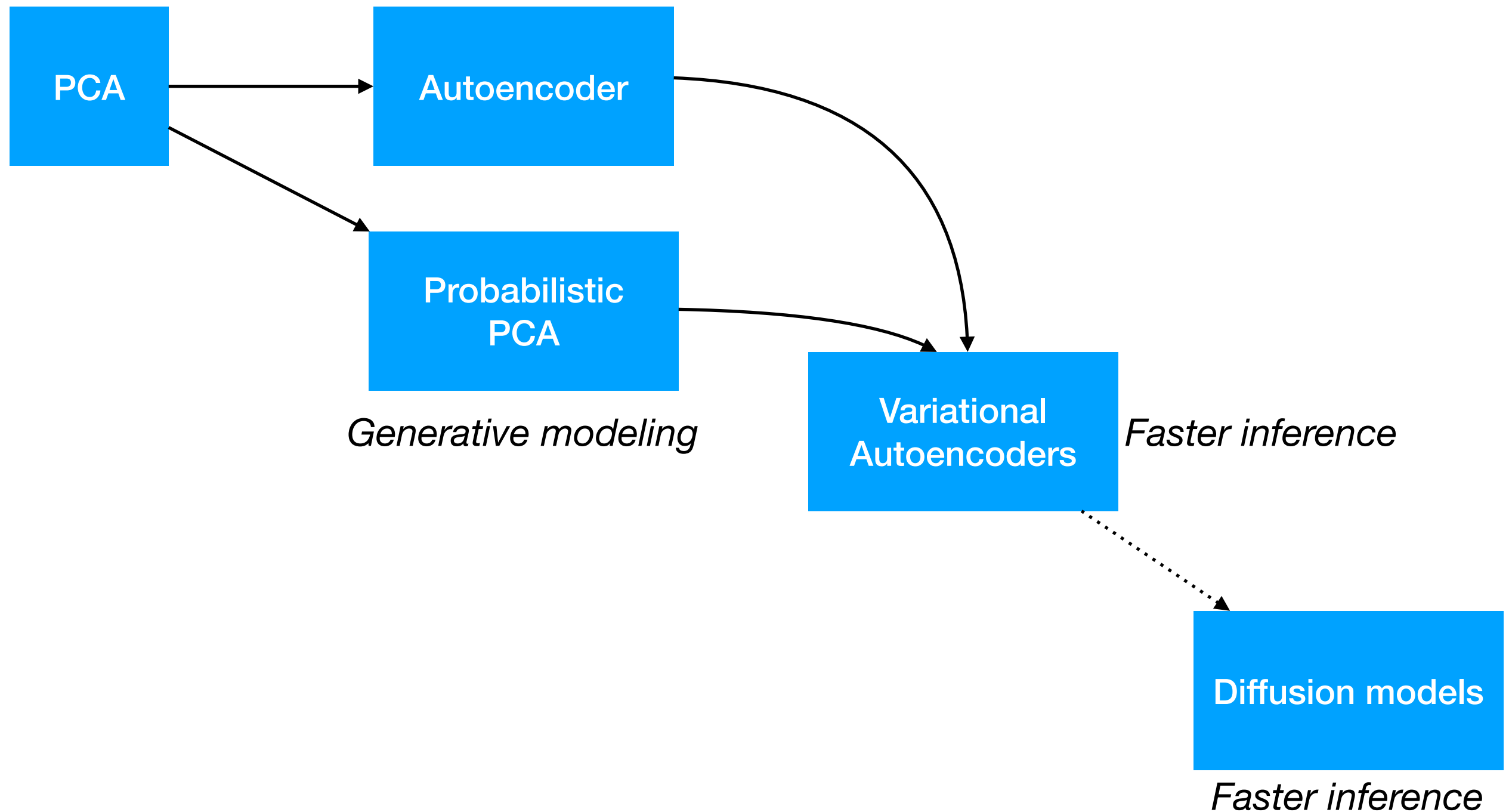
Generative modeling

*A digital Illustration of the Babel tower, 4k, detailed, trending in artstation, fantasy
vivid colors*



Outline of today's lecture

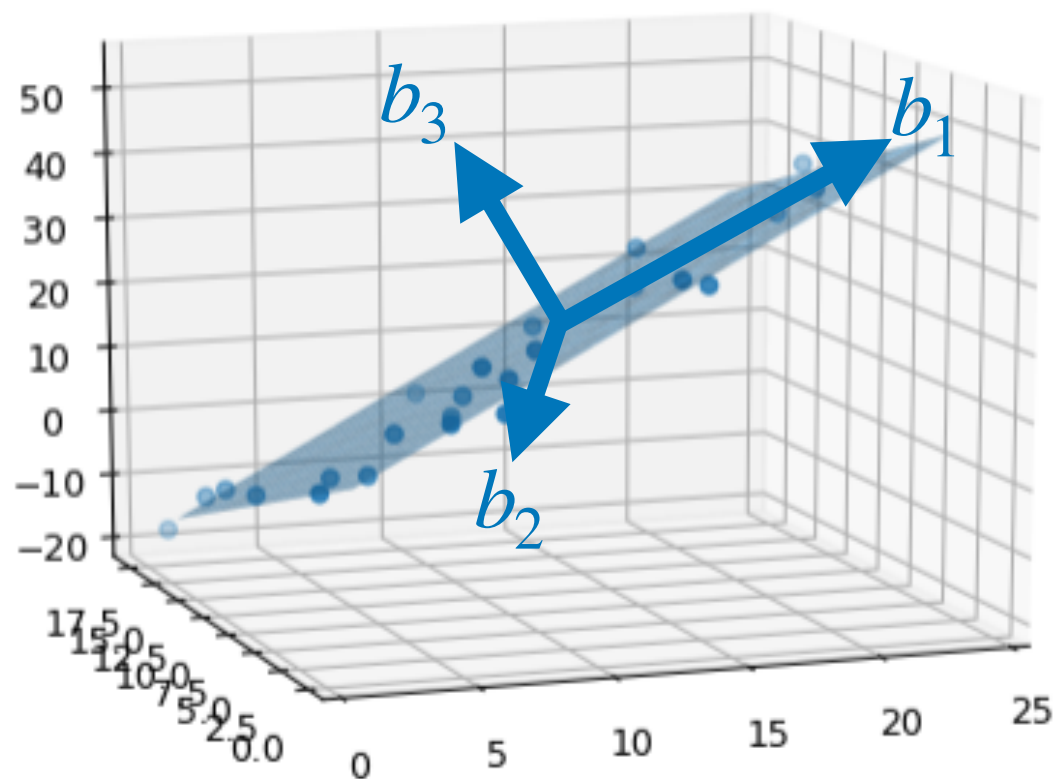
Nonlinear dimension reduction



Lecture Outline

- **Some linear algebra review**
- PCA
 - Minimum reconstruction error perspective
- Probabilistic perspective: PPCA
 - Solving the optimality condition for PPCA

Basis vectors



Standard basis:

$$\{(0,0,1), (0,1,0), (1,0,0)\}$$

Basis (in general):

$\{b_1, b_2, b_3\}$ where $b_j^\top b_k = 0$ for all $j \neq k$ “Linearly independent”

Orthonormal Basis:

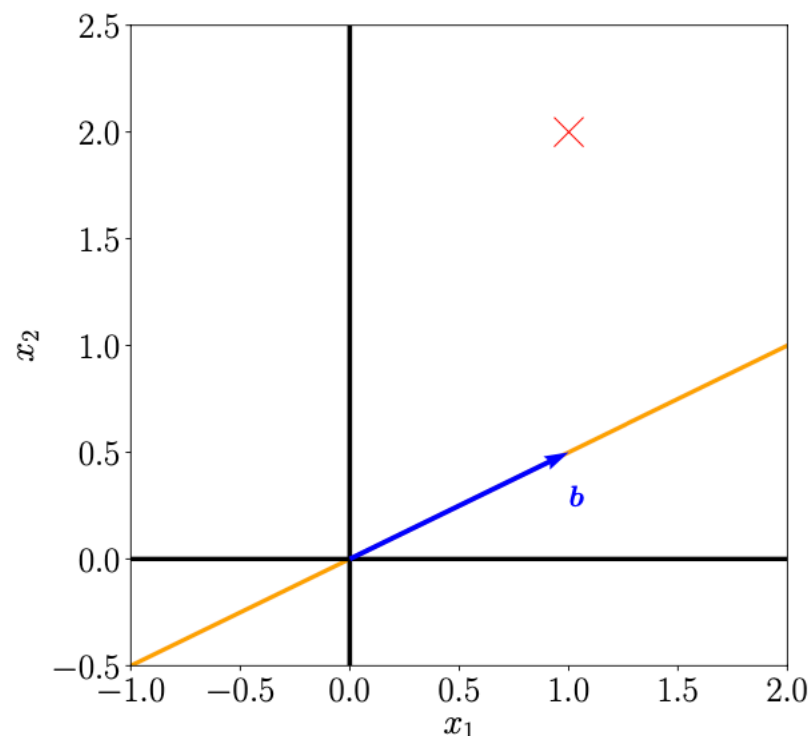
$\{b_1, b_2, b_3\}$ where

$$\|b_j\|_2 = 1 \text{ for all } j = 1, 2, 3$$

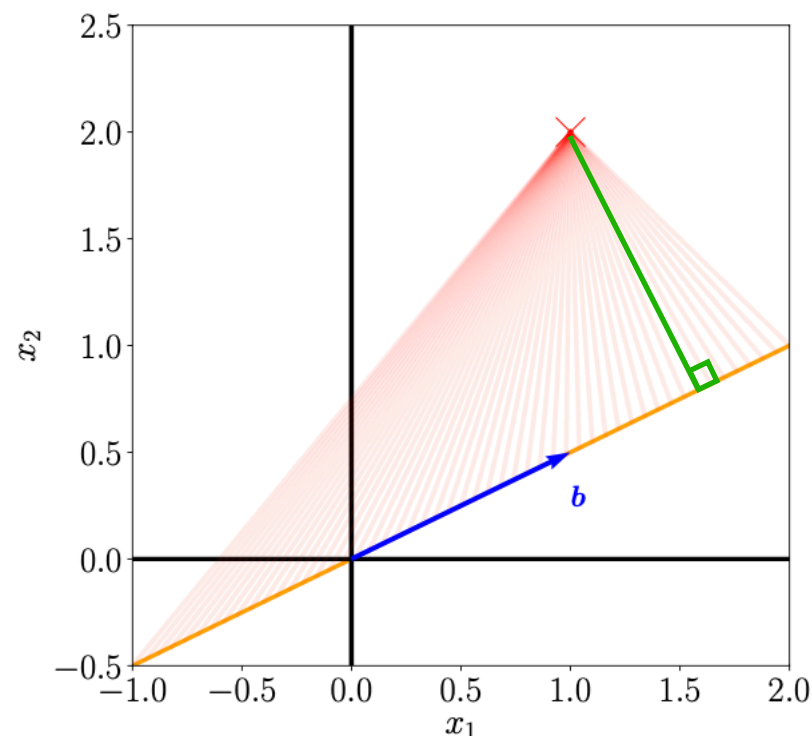
$$b_j^\top b_k = 0 \text{ for all } j \neq k$$

Projections

- The **projection** of a point x onto a vector b , denoted $\tilde{x}$, is the point that is closest to x along the vector b in terms of the Euclidean distance, i.e.
 - $\min_{\tilde{x}} \|\tilde{x} - x\|_2^2$ where $\tilde{x} = zb$ for some constant z
 - Equivalently, $\min_z \|zb - x\|_2^2$
 - Solution: $\tilde{x} = bb^\top x$, which is the point such that $x - \tilde{x}$ is **orthogonal** to b .
- Extension to multiple points X and subspace spanned by the columns of B : The projection of X onto $\text{span}(B)$ is $\tilde{X} = XBB^\top$.



(a) Setting.



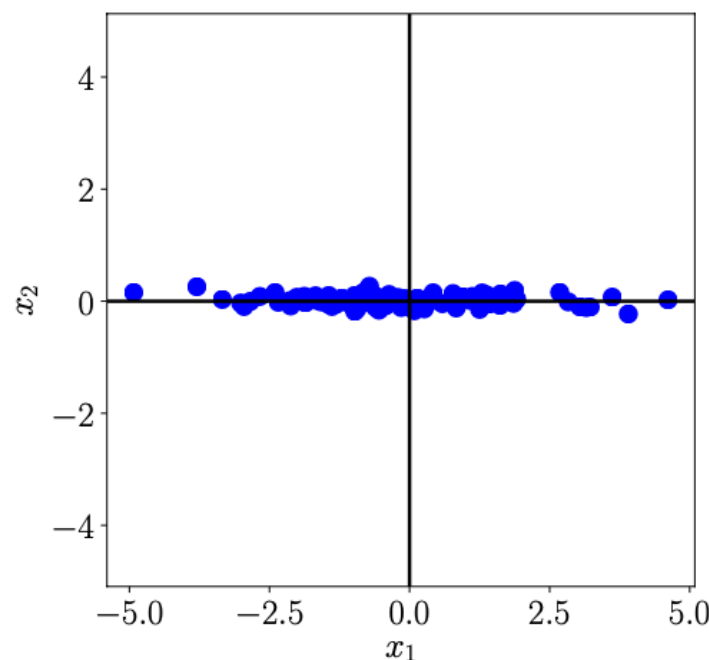
(b) Differences $x - \tilde{x}_i$ for 50 different $\tilde{x}_i$ are shown by the red lines.

Lecture Outline

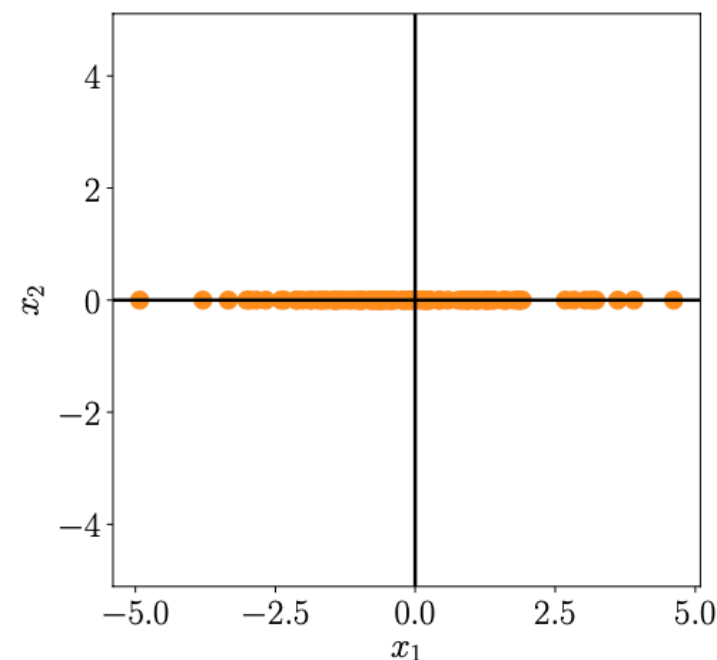
- Some linear algebra review
- **PCA**
 - Minimum reconstruction error perspective
- Probabilistic perspective: PPCA
 - Solving the optimality condition for PPCA

Principal Components Analysis (PCA)

- PCA is used for **dimension reduction** and data visualization. The central idea of PCA is that high-dimensional data is often redundant and possesses an intrinsic lower-dimensional structure.
- Variants of PCA can also be used as a **generative model**.



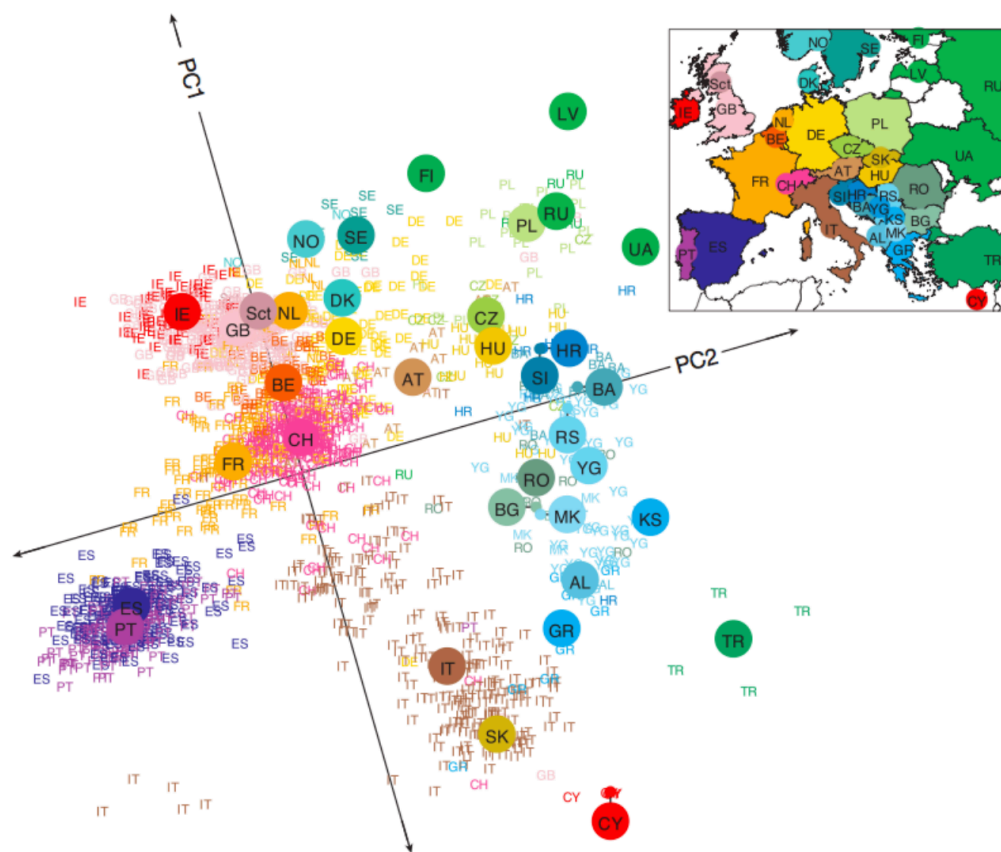
(a) Dataset with x_1 and x_2 coordinates.



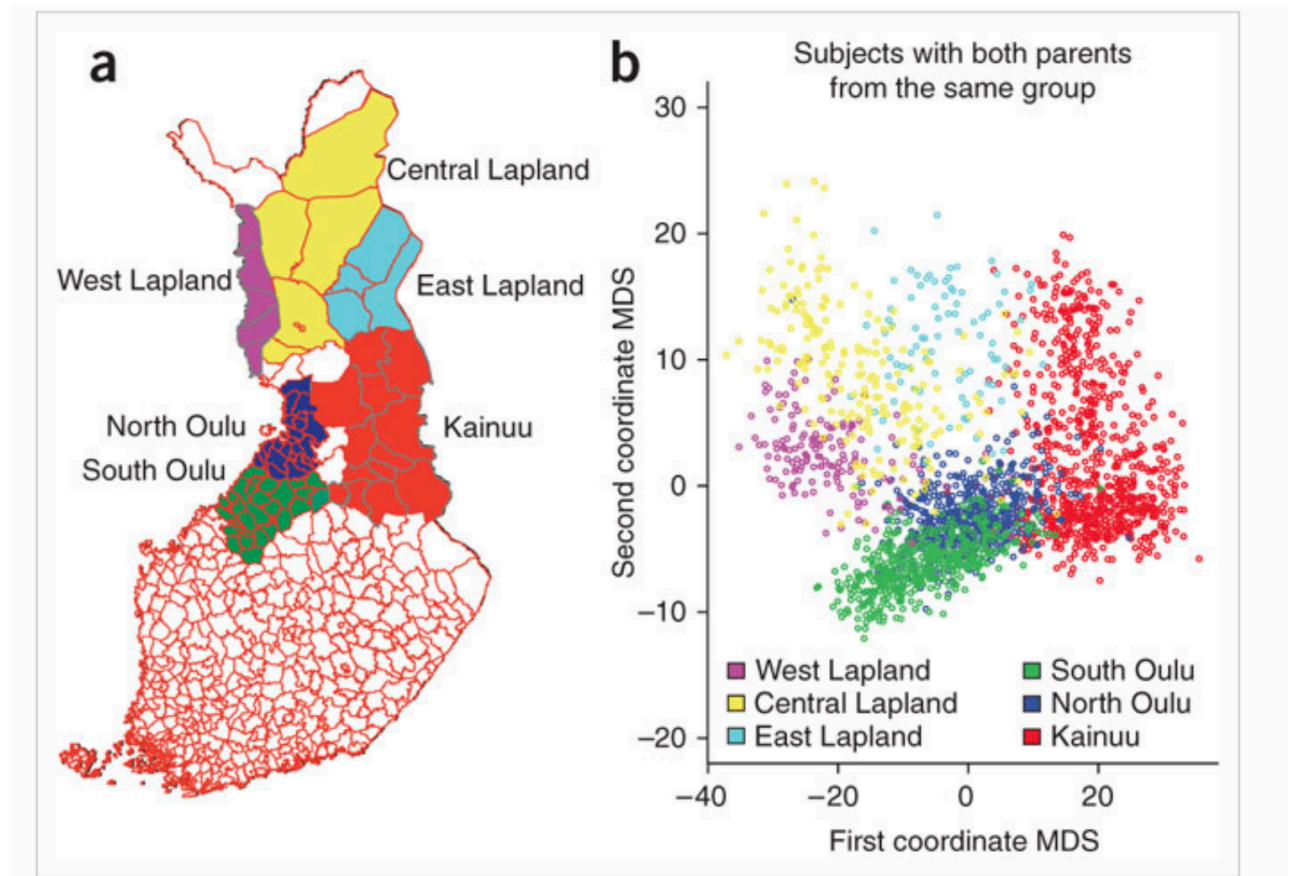
(b) Compressed dataset where only the x_1 coordinate is relevant.

Example: PCA in genetics

- PCA applied to genotype data can be used to calculate principal components (PCs) that explain differences among the sample individuals in the genetic data



Novembre et al., 2008

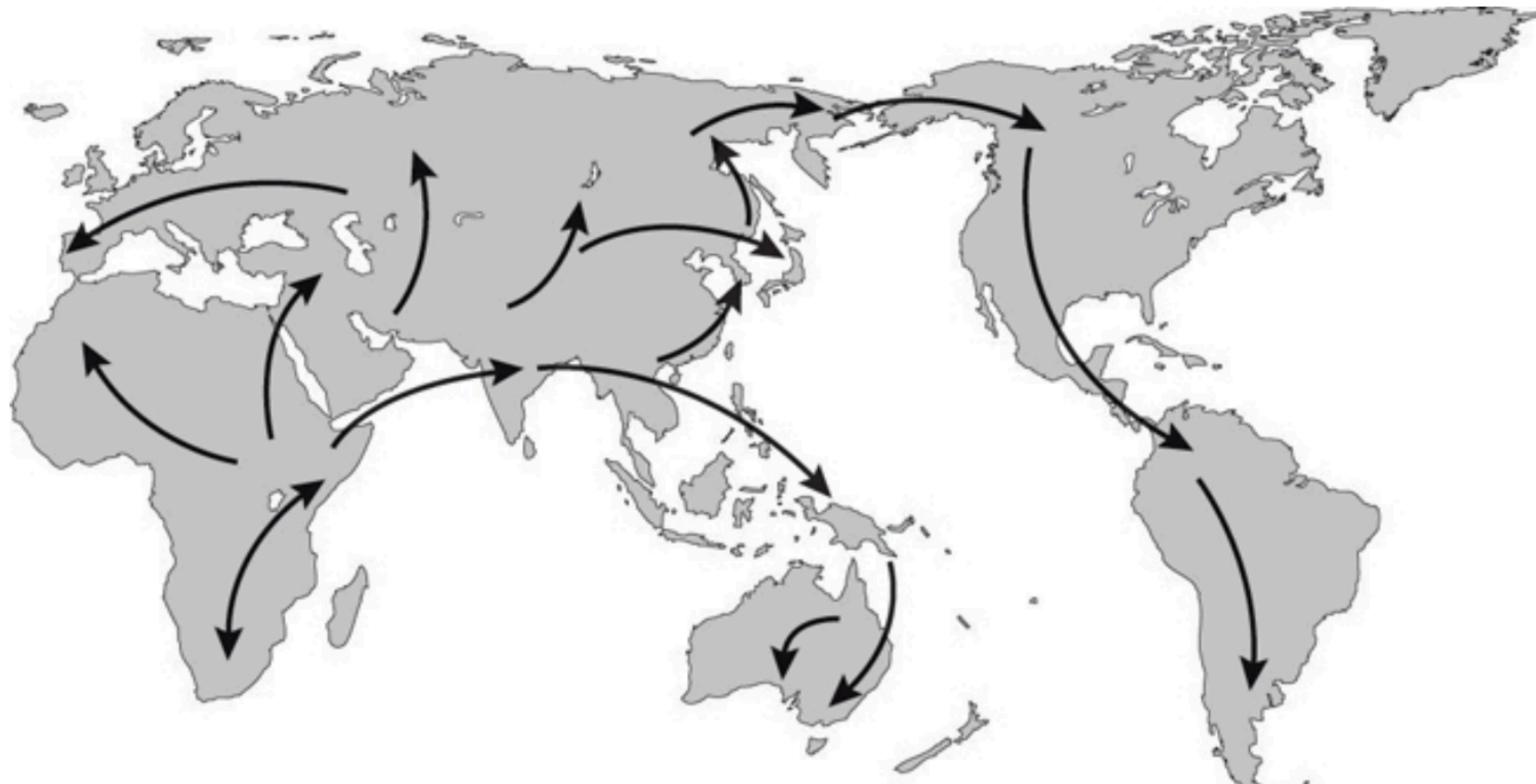


Sabatti et al., 2009

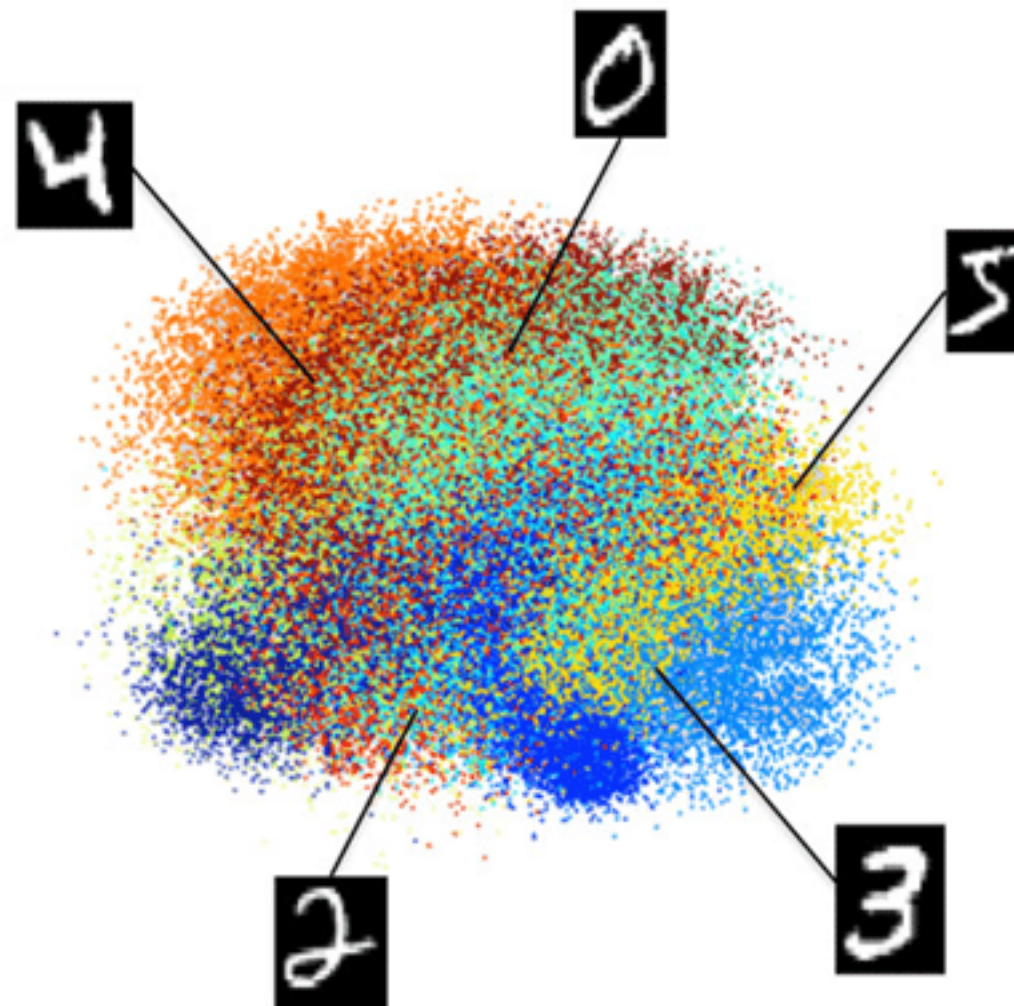
Why do the top PCs reflect geographic distance?

Example: PCA in genetics

- It is thought that the top PCs are continuous representations of genetic ancestry.

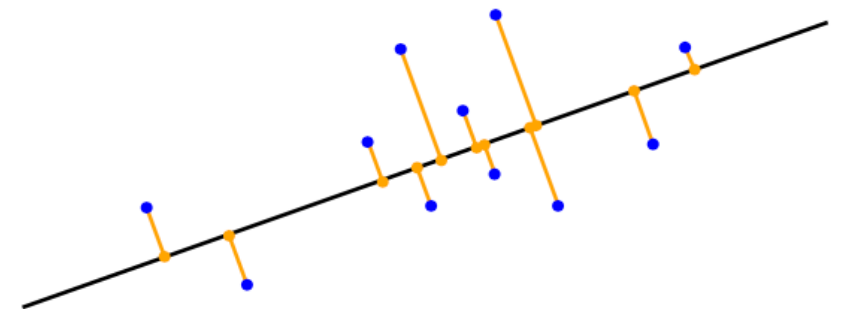
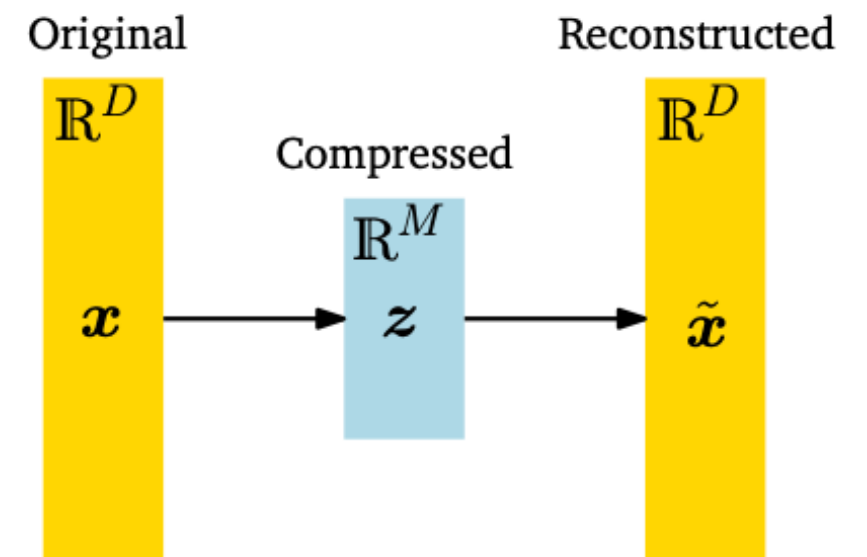


Another PCA example



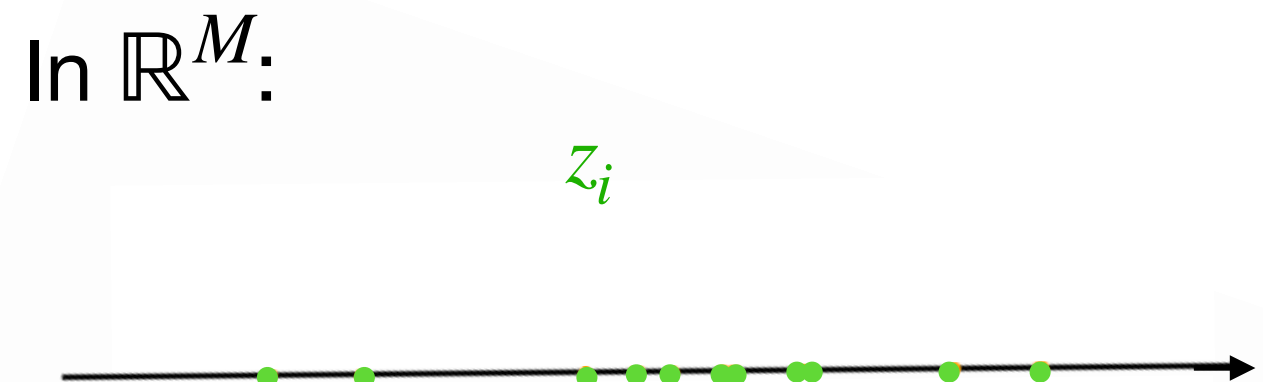
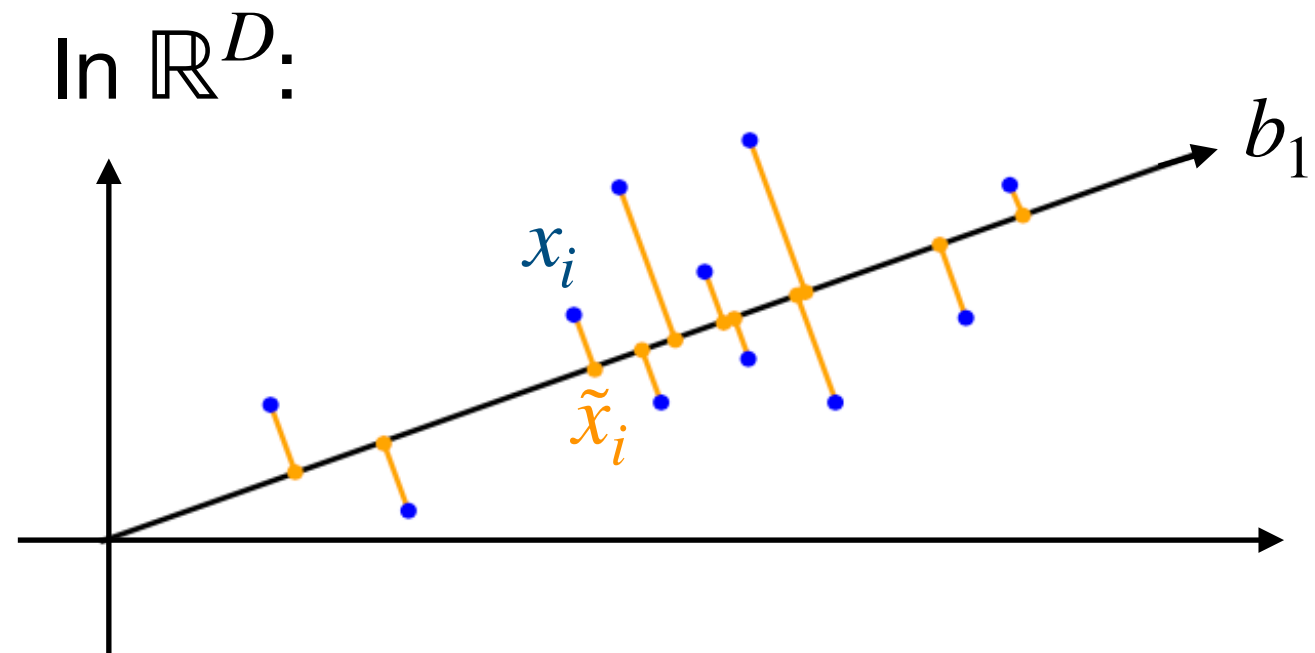
PCA

- Given data points $\{x_1, \dots, x_n\}$ where $x_i \in \mathbb{R}^D$.
- We aim to come up with a lower dimensional representation $\{z_1, \dots, z_n\}$ where $z_i \in \mathbb{R}^M$.
- How should we construct this lower dimensional representation?
 - **Maximum variation:** Find a lower-dimensional representation that captures the largest amount of variation in the data.
 - **Minimum reconstruction error:** Find a lower-dimensional representation z that minimizes the MSE between the reconstructed data based on z and the original data.
 - **Probabilistic Perspective:** Suppose the observed data is generated from a probabilistic model with latent random variable Z_i , and use MLE.



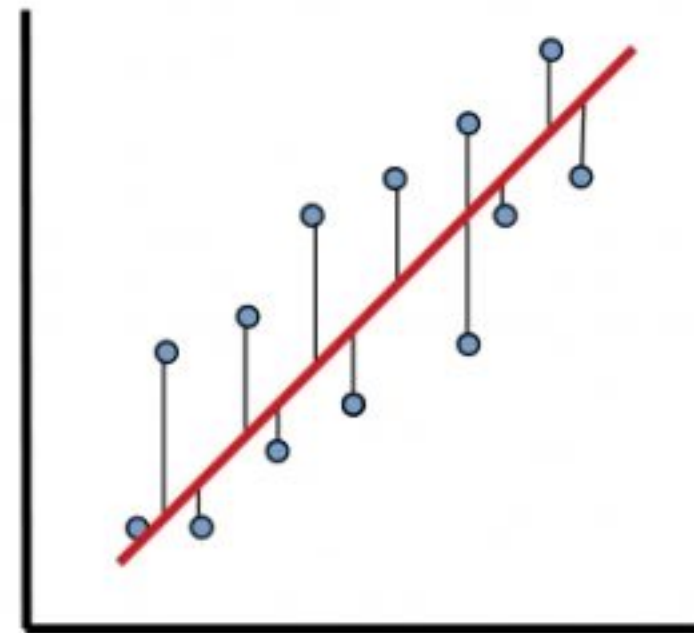
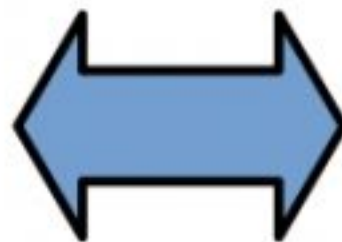
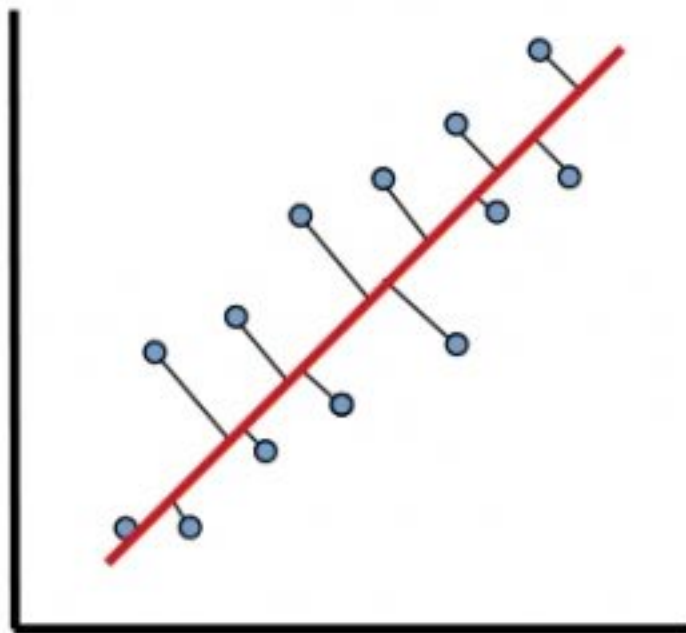
PCA

- The original data is denoted $x_i \in \mathbb{R}^D$.
- The lower-dimensional representation is denoted $z_i \in \mathbb{R}^M$.
 - z_i represents coordinates in the M -dimensional subspace of $\mathbb{R}^D$ spanned by **basis vectors** $\{b_1, \dots, b_M\}$.
- The reconstructed data is denoted $\tilde{x}_i \in \mathbb{R}^D$.
 - In PCA, the learned M -dimensional representation is the one for which the reconstructed data $\tilde{x}_i$ is a **projection** of the original data x_i in the subspace spanned by $\{b_1, \dots, b_M\}$.



Quiz

- **Q:** What's the difference between linear regression vs PCA?



Minimizing the reconstruction error

- How can we formulate the following task into an optimization problem?
 - “Find a lower-dimensional representation z that minimizes the MSE between the reconstructed data based on z and the original data x .”
- We represent the M -dimensional subspace by some orthonormal basis $\{b_1, \dots, b_M\}$.
- The reconstructed observation for basis $\{b_1, \dots, b_M\}$ and lower-dimensional representation z is defined as
$$\tilde{x} = Bz = \sum_{j=1}^M z_j b_j.$$
- Define **reconstruction error** of x by $\tilde{x}$ by the squared Euclidean distance, i.e. $\|x - \tilde{x}\|_2^2$.

$$\min_{z,b} \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^M z_{ij} b_j - x_i \right)^2$$

s.t. $\|b_j\|_2^2 = 1 \forall j = 1, \dots, m$ **Orthonormal**

$b_j^\top b_k = 0 \forall j \neq k$ **Orthogonal**

Equivalently:

$$\min_{B,Z} \|\tilde{X} - X\|_2^2$$

s.t. $\tilde{X} = ZB^\top$ and $B^\top B = I$

which is also equivalent to:

$$\min_{B,Z} \|ZB^\top - X\|_2^2$$

s.t. $B^\top B = I$

Minimizing the reconstruction error

$$\min_{z,b} \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^M z_{ij} b_j - x_i \right)^2$$
$$\text{s.t. } \|b_j\|_2^2 = 1 \forall j = 1, \dots, m$$
$$b_j^\top b_k = 0 \forall j \neq k$$

Equivalently:

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which is also equivalent to:

$$\min_{B,Z} \|ZB^\top - X\|_2^2$$
$$\text{s.t. } B^\top B = I$$

- The constrained optimization problem in PCA is **not** convex.
- Nevertheless, it is one of the few non-convex optimization problems with a beautiful, closed-form solution!
- We derive this in two steps:
 1. Find optimal coordinates z for a given orthonormal basis $B \in \mathbb{R}^{D \times M}$.
 2. Find the optimal orthonormal basis B .
- Throughout, we'll suppose the data is centered.

Step 1: Find z given B

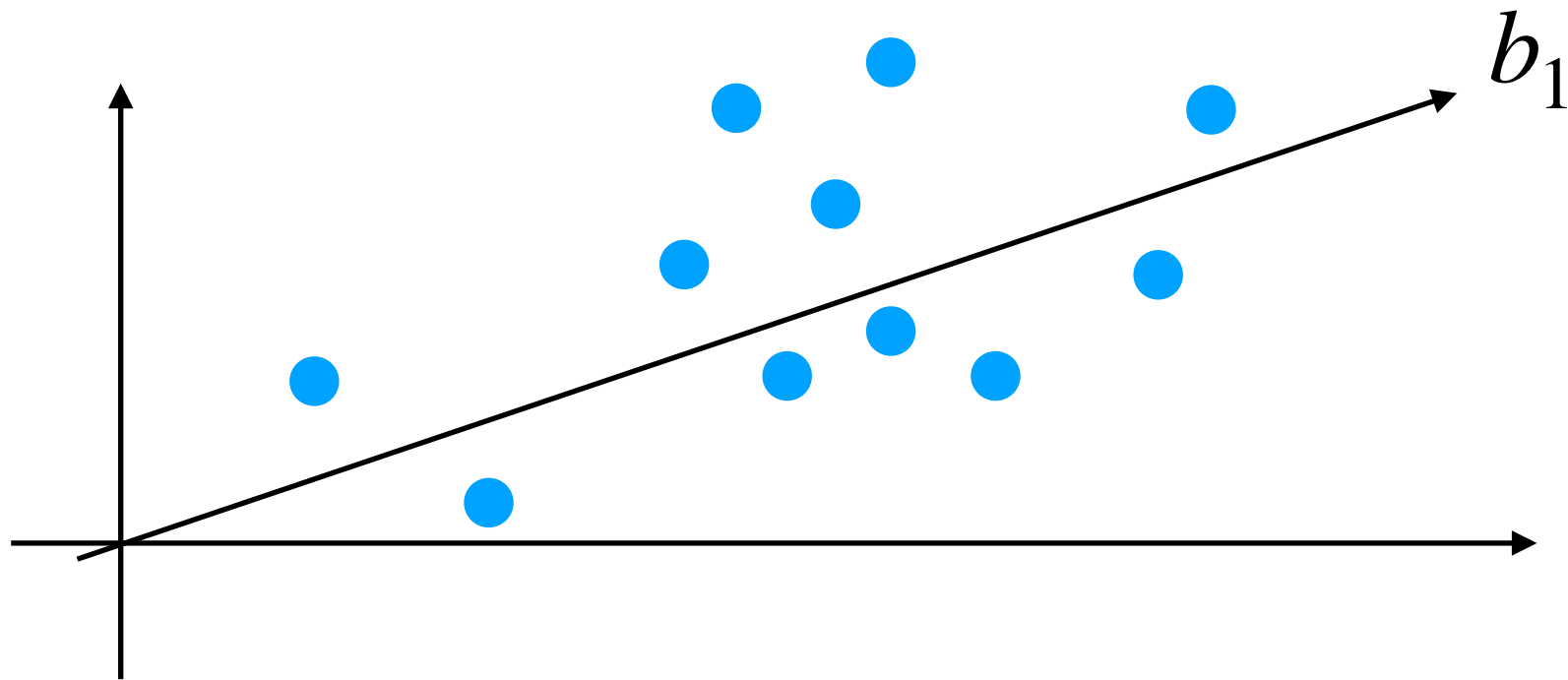
Original problem:

$$\begin{aligned} \min_{B, Z} & \|ZB^\top - X\|_2^2 \\ \text{s.t. } & B^\top B = I \end{aligned}$$

Simplified problem:

- Suppose B is fixed.

$$\min_Z \|ZB^\top - X\|_2^2$$



Step 1: Find z given B

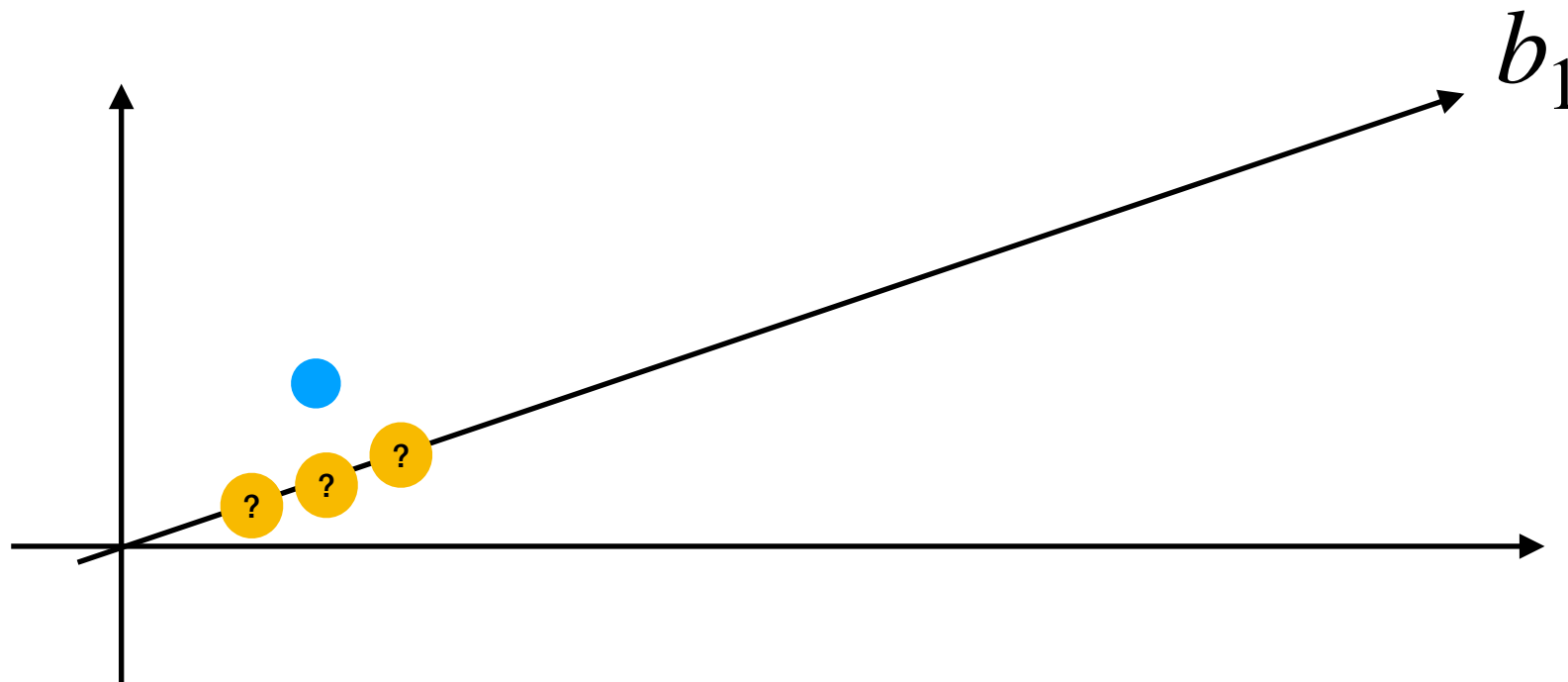
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Simplified problem:

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$$\min_Z \|ZB^\top - X\|_2^2$$

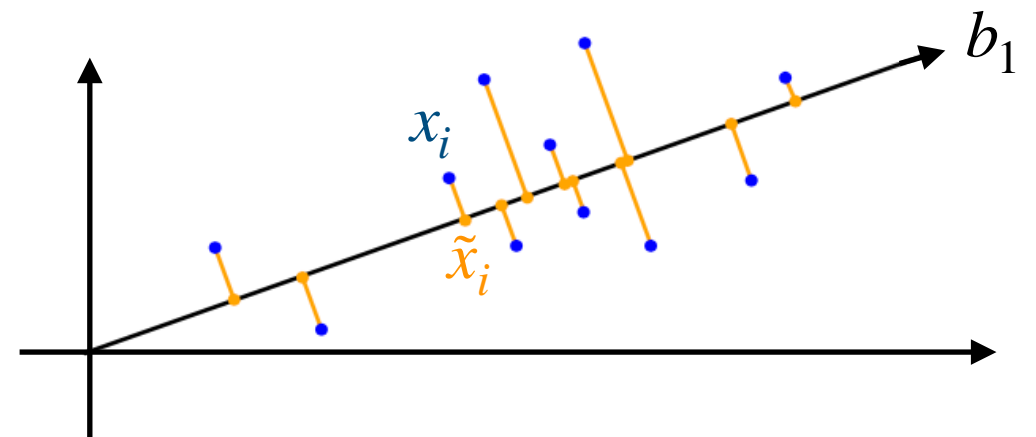
- Taking the gradient with respect to Z , we have

$$\begin{aligned} \nabla_Z \|ZB^\top - X\|_2^2 &= 2(ZB^\top - X)B \\ &= -2XB + 2ZB^\top B \\ &= -2XB + 2Z \end{aligned}$$

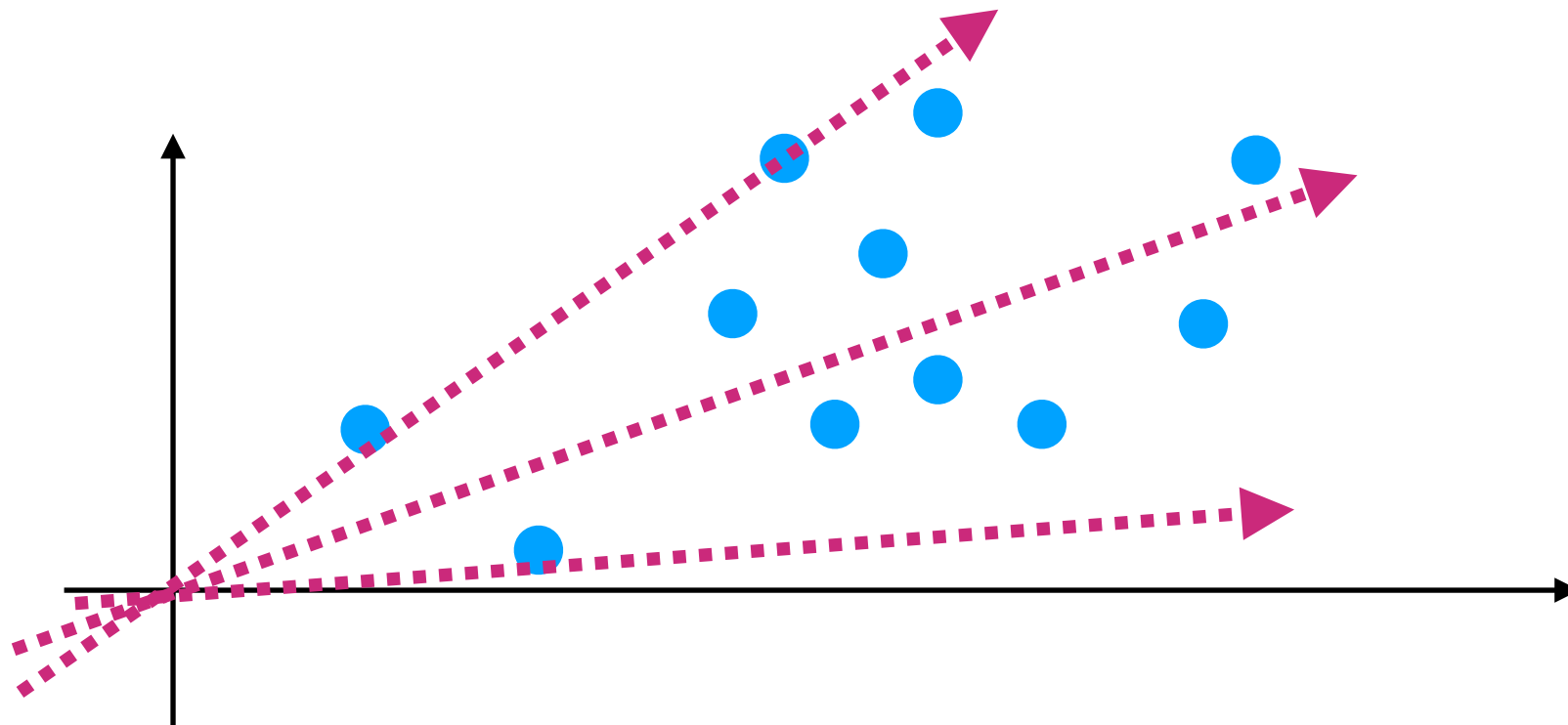
- Setting the gradient to equal to zero, we have $Z = XB$.

➡ The reconstructed points

$\tilde{X} = ZB^\top = XBB^\top$ are the **orthogonal projection** of X onto the subspace spanned by B .



Step 2: Find B



Step 2: Find B

Original problem:

$$\begin{aligned} \min_{B, Z} & \|ZB^\top - X\|_2^2 \\ \text{s.t. } & B^\top B = I \end{aligned}$$

Plug in results from Step 1:

- $Z = XB$

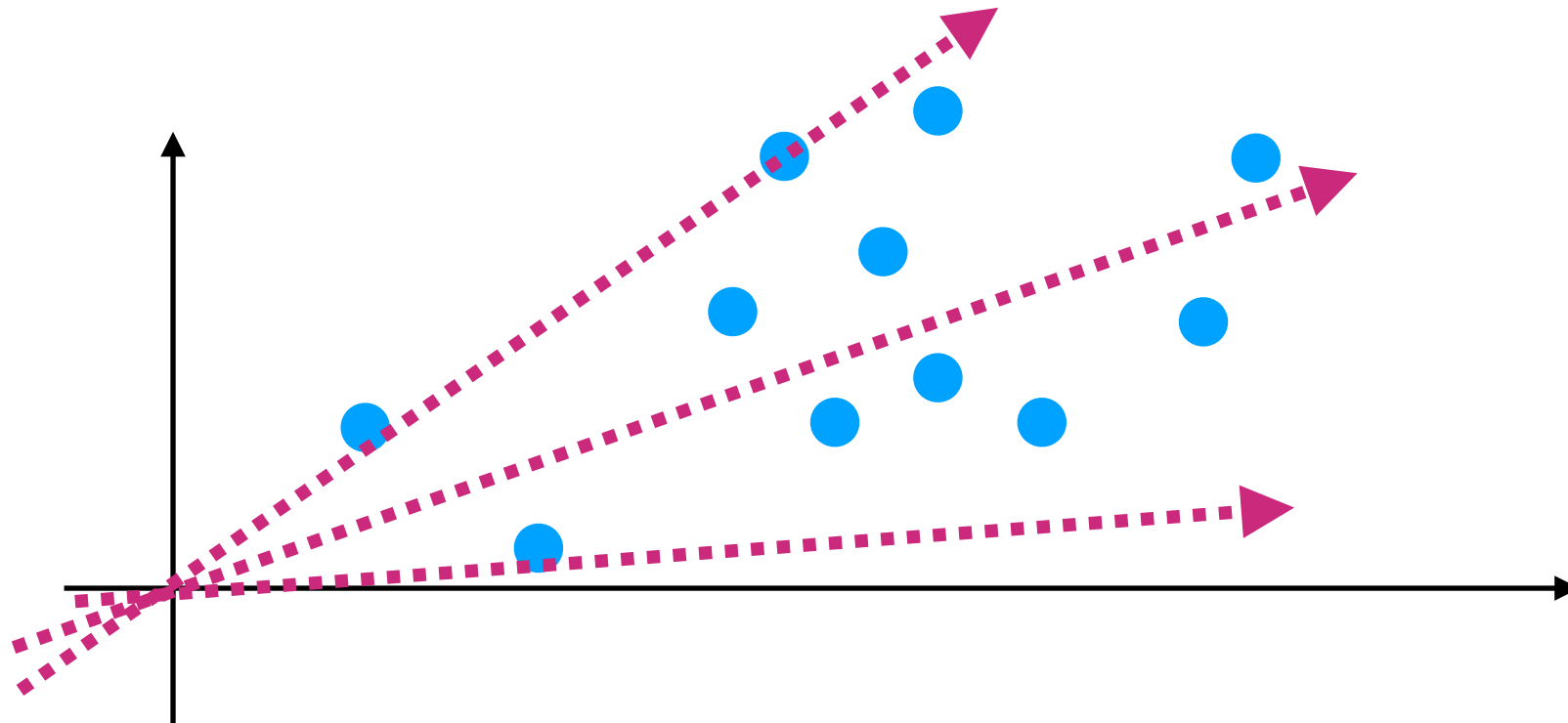
$$\begin{aligned} \min_B & \|XBB^\top - X\|_2^2 \\ \text{s.t. } & B^\top B = I \end{aligned}$$

$$\begin{aligned} \min_B \| (I - BB^\top)X \|_2^2 & \Leftrightarrow \min_B \sum_{i=1}^n x_i^\top (I - BB^\top)^\top (I - BB^\top) x_i \\ & \Leftrightarrow \min_B \sum_{i=1}^n x_i^\top (I - 2BB^\top + BB^\top BB^\top) x_i \\ & \Leftrightarrow \min_B \sum_{i=1}^n x_i^\top (I - 2BB^\top + BB^\top) x_i \\ & \Leftrightarrow \min_B \sum_{i=1}^n x_i^\top (I - BB^\top) x_i \\ & \Leftrightarrow \min_B - \sum_{i=1}^n x_i^\top BB^\top x_i \\ & \Leftrightarrow \max_B \sum_{i=1}^n x_i^\top BB^\top x_i \\ & \Leftrightarrow \max_B \sum_{j=1}^M b_j^\top X^\top X b_j \end{aligned}$$

B are the M directions that capture the maximum variation in the data

Step 2: Find B

If we set $M = 1$, what is the direction that captures the maximum variation in the data?



Step 2: Find the optimal basis

It turns out we can solve $\max_B \sum_{j=1}^M b_j^\top X^\top X b_j$ for orthonormal basis B sequentially!

1. $\max_{b_1} b_1^\top X^\top X b_1$ such that $\|b_1\|_2^2 = 1$

2. $\max_{b_2} b_2^\top X^\top X b_2$ such that $\|b_2\|_2^2 = 1, b_2^\top b_1 = 0$

3. ...

Step 2: Find the first PC

- **Q:** How can we solve for the first PC?

$$\max_{b_1} b_1^\top X^\top X b_1 \text{ such that } \|b_1\|_2^2 = 1$$

- **A:** Gradient descent
- **B:** Get a closed-form solution by setting the gradient of the objective function $b_1^\top X^\top X b_1$ with respect to b_1 to zero.
- **C:** You need some solver for a constrained optimization problem... which we haven't learned yet.

Principal Components Analysis (PCA)

For $k = 1, \dots, K$

Solve $\max_{b_k: \|b_k\|_2=1} \underbrace{b_k^\top X^\top X b_k}_{\text{Variance of } X \text{ along axis defined by } b_k}$ subject to $b_k^\top b_j = 0 \quad \forall j < k$

Define the k th PC as $Z_k = Xb_k$.

Turns out this is simply doing a Singular Value Decomposition!

(Descending order along diagonal)

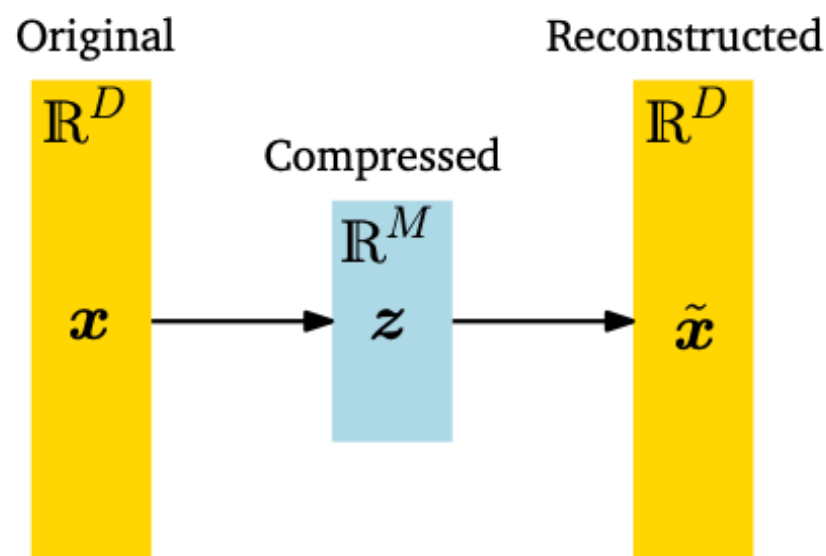
Perform Singular Value Decomposition $X = UDB^\top$

The PC scores are given by $Z_1, \dots, Z_K$ where $Z = XB$.

The PC loadings are $b_1, \dots, b_K$.

Extending PCA

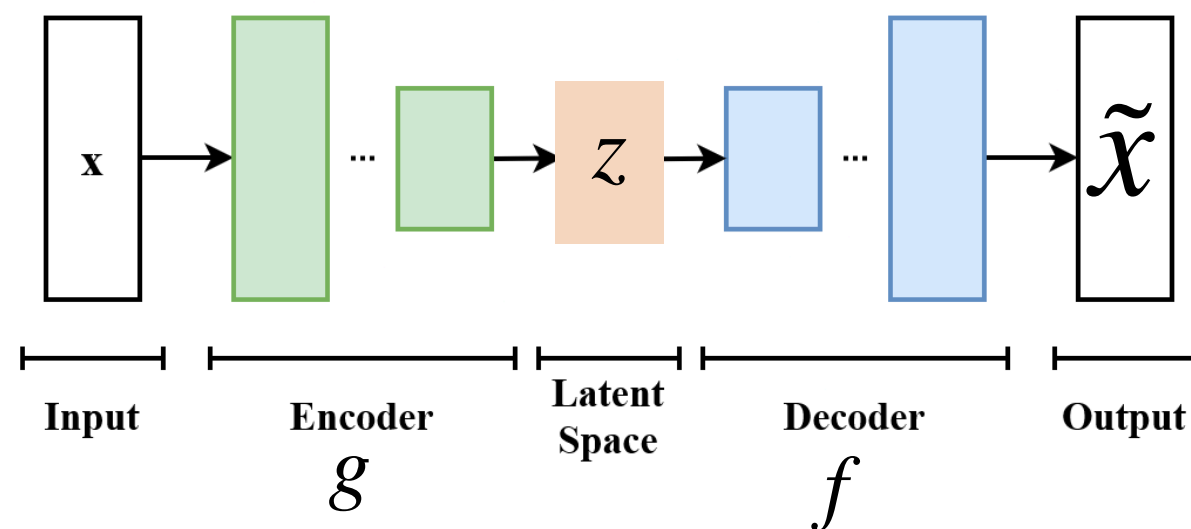
PCA



$$\min_{B, Z} \|XBB^\top - X\|_2^2$$

$$\text{s.t. } B^\top B = I$$

Autoencoders



$$\min_{f, g} \frac{1}{n} \sum_{i=1}^n \|f(g(x)) - x\|^2$$

such that

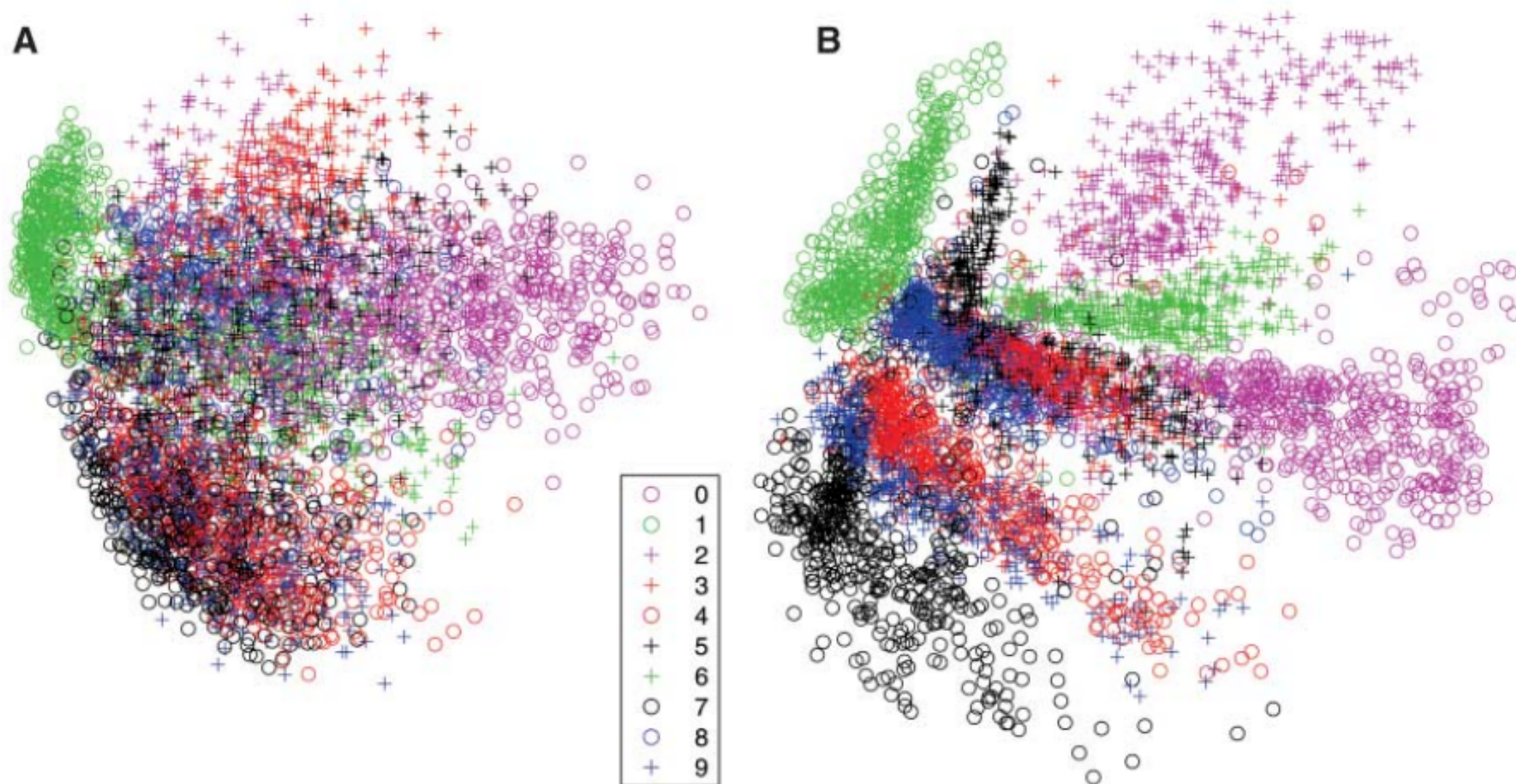
$$f: \mathbb{R}^M \mapsto \mathbb{R}^D \text{ and}$$

$$g: \mathbb{R}^D \mapsto \mathbb{R}^M$$

PCA vs Autoencoders



Fig. 3. (A) The two-dimensional codes for 500 digits of each class produced by taking the first two principal components of all 60,000 training images. (B) The two-dimensional codes found by a 784-1000-500-250-2 autoencoder. For an alternative visualization, see (8).

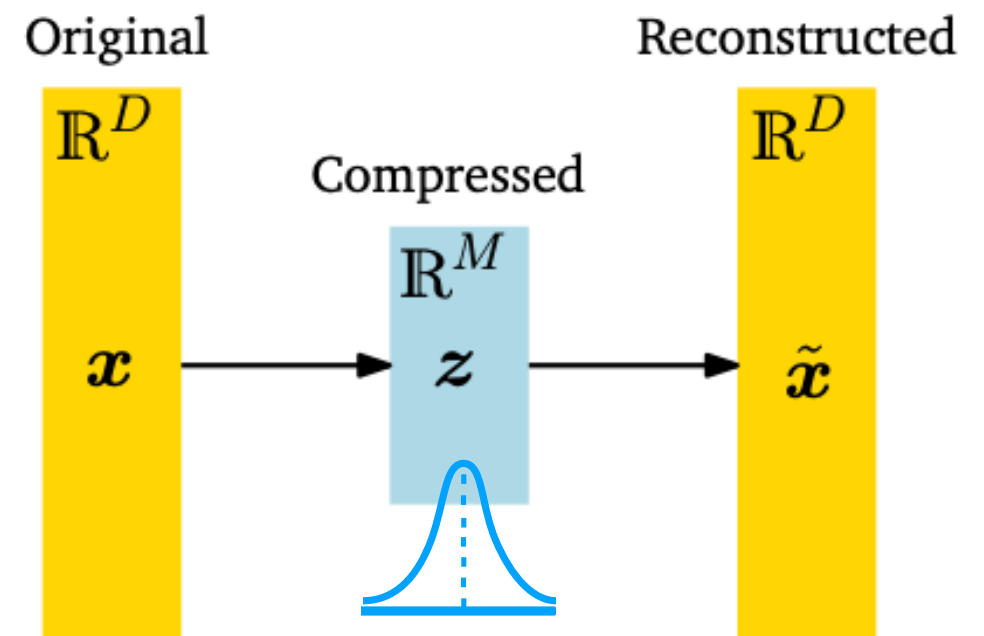


Lecture Outline

- Some linear algebra review
- PCA
 - Minimum reconstruction error perspective
- **Probabilistic perspective: PPCA**
 - Solving the optimality condition for PPCA

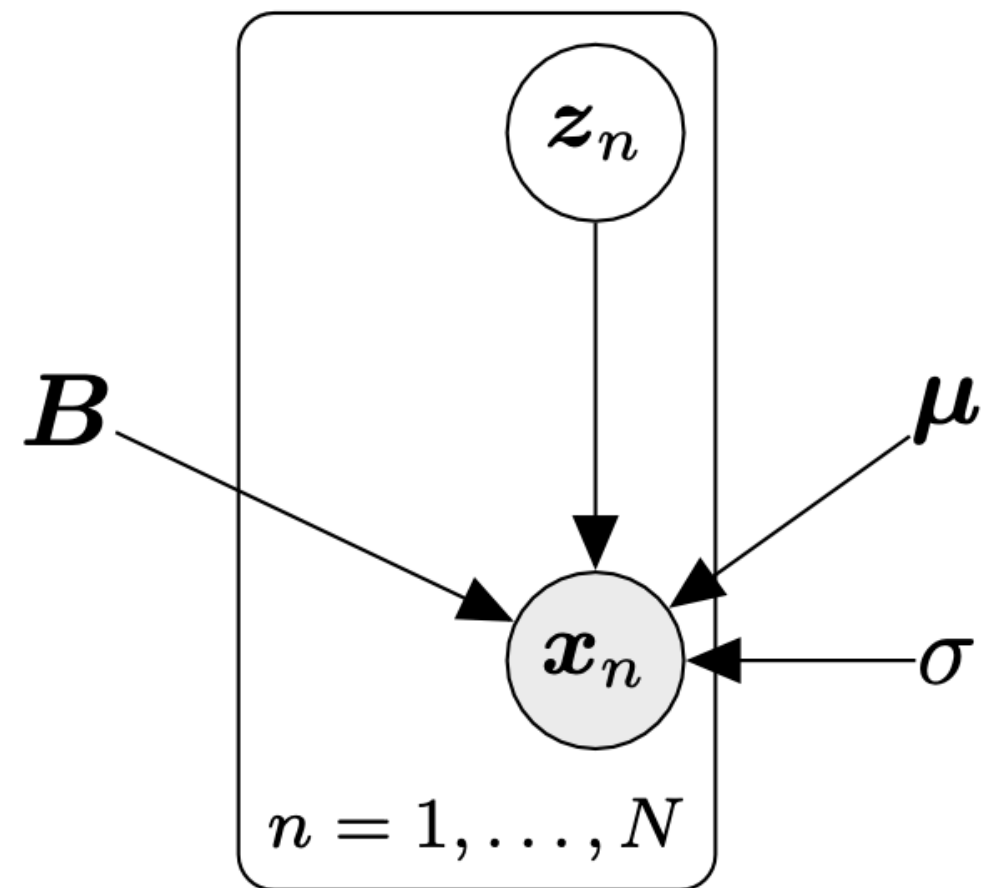
Probabilistic PCA

- **Probabilistic PCA** extends ordinary PCA by defining a probabilistic model of the data.
- Benefits:
 - It's a **generative model**
 - Provides uncertainty quantification
 - Deals with missing data
 - Can integrate prior knowledge via a Bayesian formulation



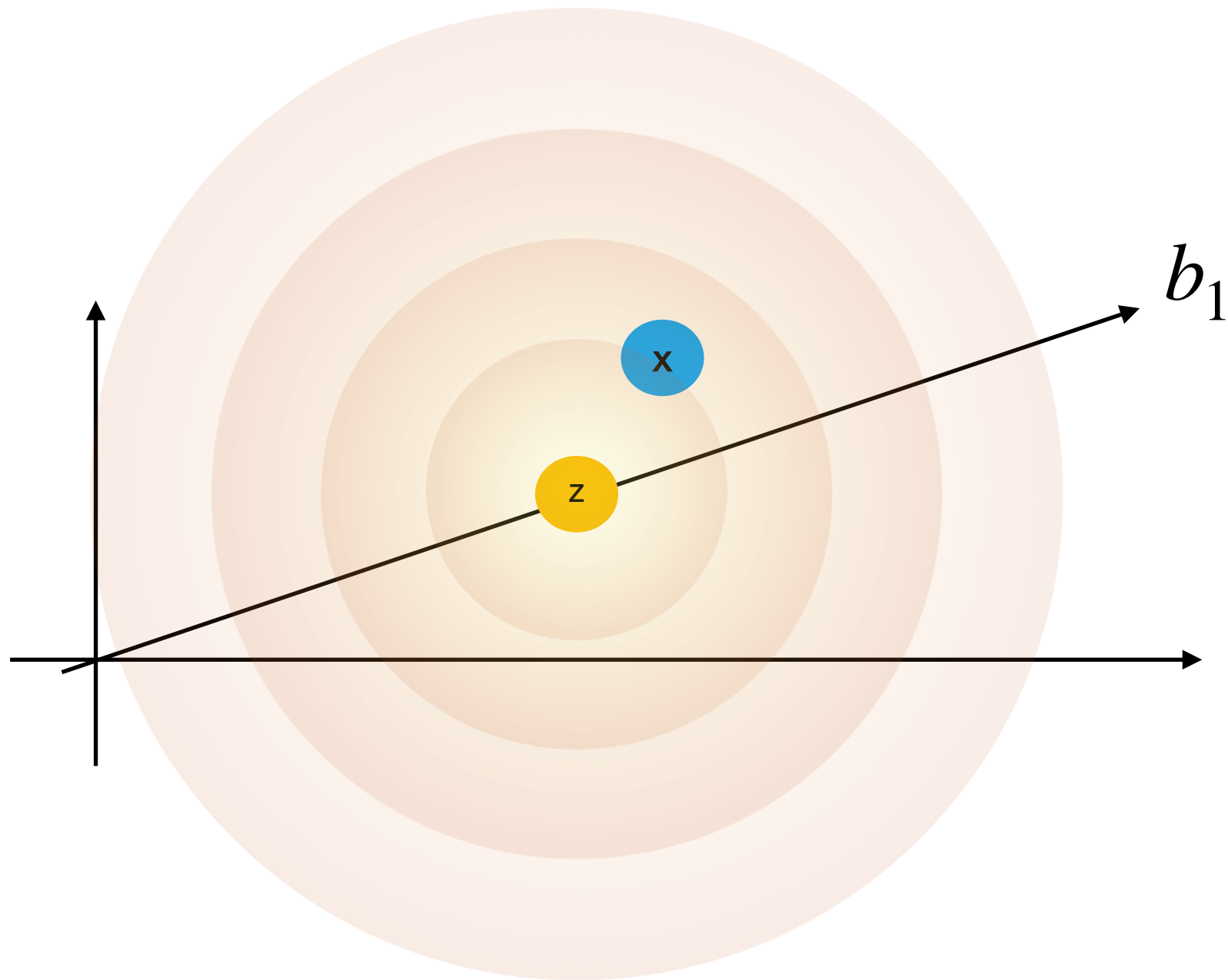
Probabilistic PCA

- Assume a standard normal prior over $z \sim N(0, I)$.
- Suppose $x = Bz + \mu + \epsilon \in \mathbb{R}^d$ where $\epsilon \sim N(0, \sigma^2 I)$
- Equivalently,
 $x | z \sim N(Bz + \mu, \sigma^2 I)$



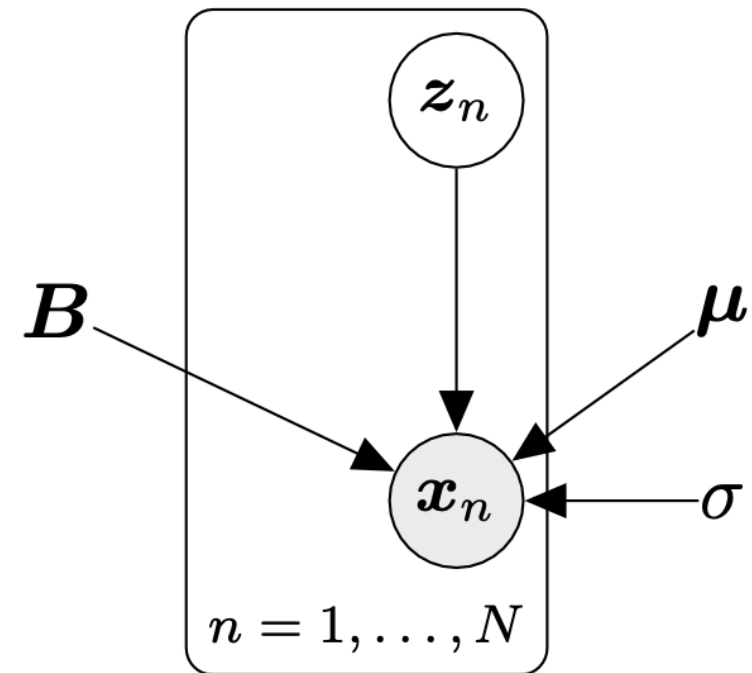
What are the unknown parameters in PPCA?

How PPCA models the data



Generating data from PPCA

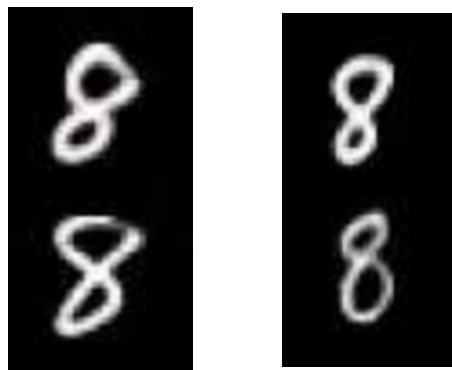
- Once you fit PPCA, we can generate new data!
 1. Sample new z and ϵ .
 2. Apply transformation
 $x = Bz + \mu + \epsilon$.



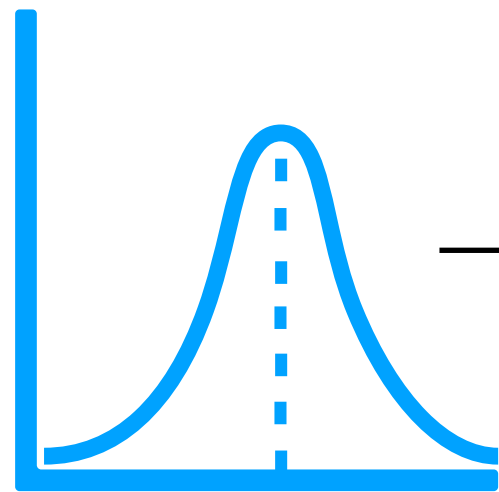
Generating data from PPCA

Step 1: Fit a PPCA model to the data.

Estimate B , μ , and σ



Step 2: From pure noise, we can generate new images of the digit 8!



$$Z \sim N(0, I)$$

$$\epsilon \sim N(0, \sigma^2 I)$$

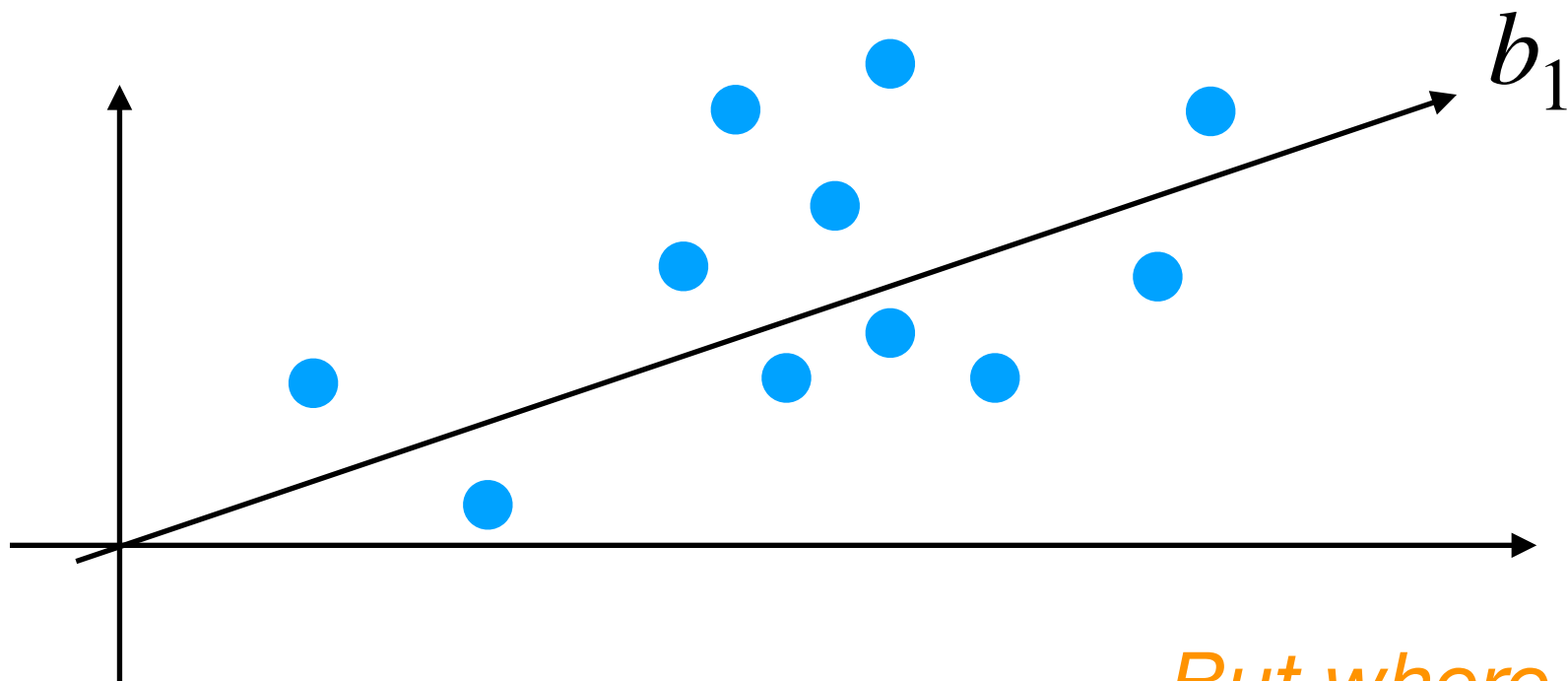
$$x = Bz + \mu + \epsilon$$



Why is PPCA hard?

- PPCA can be solved via **maximum likelihood estimation**.
- *But what is the likelihood of the observed data?*
- The likelihood of the **complete** data, *including the latent variables z* , is

$$\prod_{i=1}^n p(x_i, z_i) = \prod_{i=1}^n p(x_i | z_i) p(z_i)$$

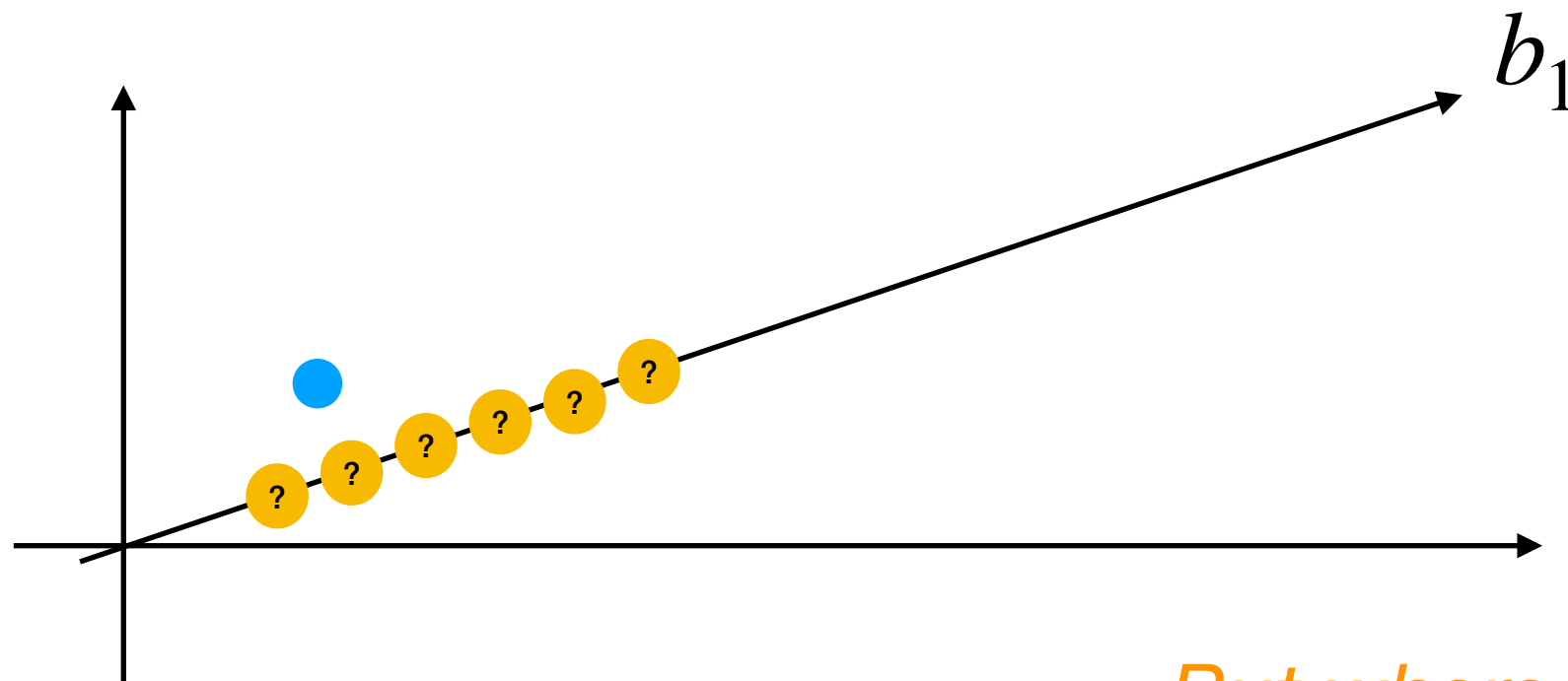


But where is Z ?!

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But where is Z ?!

Probabilistic PCA

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variables z , is
$$\prod_{i=1}^n p(x_i, z_i) = \prod_{i=1}^n p(x_i | z_i) p(z_i)$$

- The likelihood of the **observed** data (x only) is

$$\prod_{i=1}^n p(x_i) = \prod_{i=1}^n \int p(x_i | z_i) p(z_i) dz_i$$

Integrals. Yuck!

- (If we also have missing data, there are even more integrals!)

How can we solve PPCA?

- When there are latent variables, we essentially have a **missing data problem**.
- Possible approaches:
 - **Marginalization**: If you're lucky, you can marginalize out the latent variables using clever math to make the integral disappear.
 - **Expectation Maximization**: A foundational technique for estimating parameters of a model in the presence of missing data
 - **Variational Inference**: A fast approximate procedure for estimating parameters of a model in the presence of missing data
 - **Bayesian inference**: If you crank the Bayesian machinery, it will propagate uncertainty (including over the missing data) accordingly.

The marginal likelihood

- It turns out we're lucky! We *can* come up with a closed-form expression for PPCA because we've assumed the data come from particularly simple Gaussian models.
- The **marginal likelihood** of the **observed** data:

$$\begin{aligned} p(x) &= \int p(x|z)p(z)dz \\ &\propto \int \exp\left(-\frac{\|x - (Bz + \mu)\|^2}{2\sigma^2}\right) \exp\left(-\frac{\|z\|^2}{2}\right) dz \\ &\propto \exp\left(-\frac{(x - \mu)^\top (BB^\top + \sigma^2 I)^{-1} (x - \mu)}{2\sigma^2}\right) \end{aligned} \quad \textit{Tricky math!}$$
$$\Rightarrow X \sim N(\mu, BB^\top + \sigma^2 I)$$

MLE for PPCA

- Now that the likelihood of the observed data does not involve a pesky integration term, we can use our usual tools!

$$X \sim N(\mu, BB^\top + \sigma^2 I)$$

- MLE for PPCA solves the optimization problem:

$$\max_{B, \sigma^2, \mu} \sum_{i=1}^n \log p(x_i)$$

$$\iff \min_{B, \sigma^2, \mu} \sum_{i=1}^n \frac{(x_i - \mu)^\top (BB^\top + \sigma^2 I)^{-1} (x_i - \mu)}{2\sigma^2}$$

MLE for PPCA

(Aka your homework)

Taking the gradient and setting it to zero, we get a closed form solution for the MLE:

- $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$

Square root of a matrix, i.e. (Take a look at `scipy.linalg.sqrtm`)
 $A = CC \implies A^{1/2} = C$

- $\hat{B} = T(\Lambda - \hat{\sigma}^2 I)^{1/2} R$ where:

- T contains the eigenvectors of the data covariance matrix $\hat{\Sigma} = \text{Cov}(X)$ corresponding to the top M eigenvalues
- $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_M)$ is the diagonal matrix with the eigenvalues associated with those in T
- R is an arbitrary orthogonal matrix (the MLE is unique up to an arbitrary orthogonal transformation)

- $\hat{\sigma}^2 = \frac{1}{D - M} \sum_{j=M+1}^D \lambda_j$ where

- λ_j are the remaining eigenvalues.

How do we get eigenvalues and eigenvectors of a matrix, i.e. $\hat{\Sigma} = T\Lambda T^T$?

(Take a look at `scipy.linalg.eig`)

This means there are many possible solutions!

So which one would you implement in your HW?

Details are also in MML Section 10.8

Lecture Outline

- Some linear algebra review
- PCA
 - Minimum reconstruction error perspective
- Probabilistic perspective: PPCA
 - Solving the optimality condition for PPCA
 - **Expectation Maximization**

PPCA solution 2: EM approach

*J. R. Statist. Soc. B (1999)
61, Part 3, pp. 611–622*

Probabilistic principal component analysis

Michael E. Tipping and Christopher M. Bishop
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Appendix B: An EM algorithm for probabilistic principal component analysis

In the EM approach to maximizing the likelihood for PPCA, we consider the latent variables $\{\mathbf{x}_n\}$ to be ‘missing’ data and the ‘complete’ data to comprise the observations together with these latent variables. The corresponding complete-data log-likelihood is then

$$\mathcal{L}_C = \sum_{n=1}^N \ln \{p(\mathbf{t}_n, \mathbf{x}_n)\}, \quad (22)$$

where, in PPCA, from the definitions in Section 3.1,

$$p(\mathbf{t}_n, \mathbf{x}_n) = (2\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|\mathbf{t}_n - \mathbf{W}\mathbf{x}_n - \boldsymbol{\mu}\|^2}{2\sigma^2}\right) (2\pi)^{-q/2} \exp\left(-\frac{\|\mathbf{x}_n\|^2}{2}\right). \quad (23)$$

In the E-step, we take the expectation of $\mathcal{L}_C$ with respect to the distributions $p(\mathbf{x}_n|\mathbf{t}_n, \mathbf{W}, \sigma^2)$:

$$\begin{aligned} \langle \mathcal{L}_C \rangle = & -\sum_{n=1}^N \left\{ \frac{d}{2} \ln(\sigma^2) + \frac{1}{2} \text{tr}(\langle \mathbf{x}_n \mathbf{x}_n^T \rangle) + \frac{1}{2\sigma^2} (\mathbf{t}_n - \boldsymbol{\mu})^T (\mathbf{t}_n - \boldsymbol{\mu}) - \frac{1}{\sigma^2} \langle \mathbf{x}_n \rangle^T \mathbf{W}^T (\mathbf{t}_n - \boldsymbol{\mu}) \right. \\ & \left. + \frac{1}{2\sigma^2} \text{tr}(\mathbf{W}^T \mathbf{W} \langle \mathbf{x}_n \mathbf{x}_n^T \rangle) \right\}, \end{aligned} \quad (24)$$

where we have omitted terms that are independent of the model parameters and

$$\langle \mathbf{x}_n \rangle = \mathbf{M}^{-1} \mathbf{W}^T (\mathbf{t}_n - \boldsymbol{\mu}), \quad (25)$$

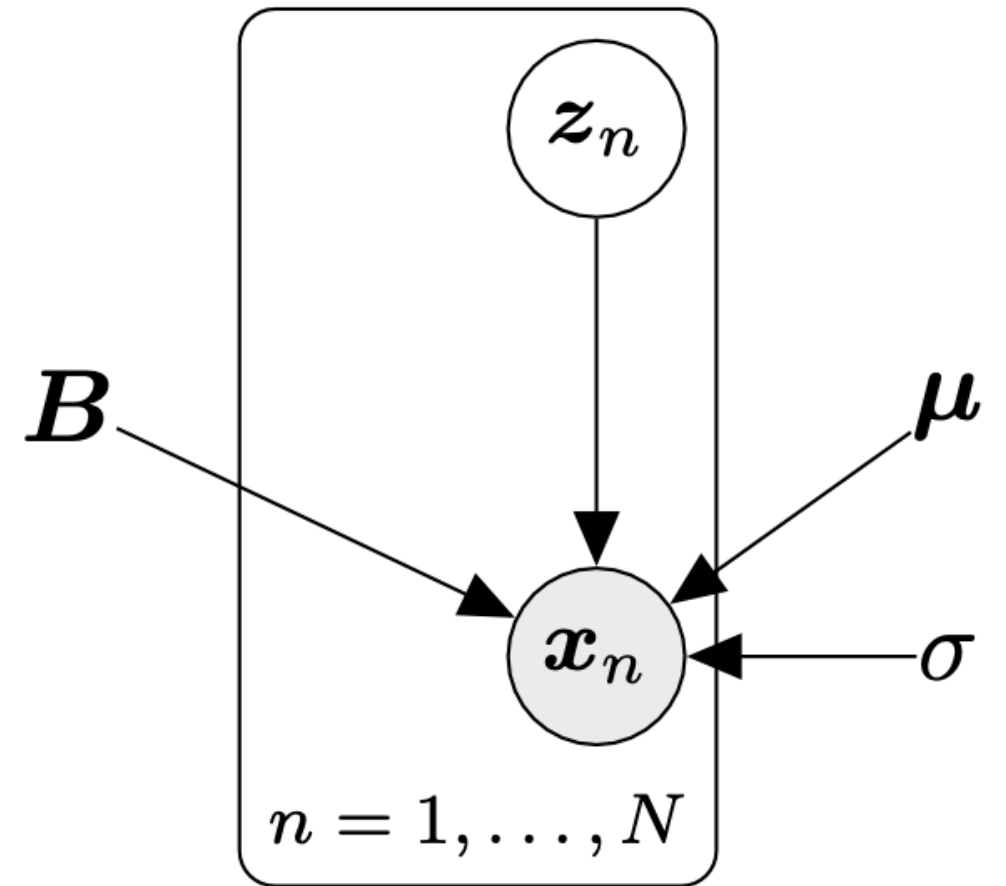
$$\langle \mathbf{x}_n \mathbf{x}_n^T \rangle = \sigma^2 \mathbf{M}^{-1} + \langle \mathbf{x}_n \rangle \langle \mathbf{x}_n \rangle^T, \quad (26)$$

in which $\mathbf{M} = \mathbf{W}^T \mathbf{W} + \sigma^2 \mathbf{I}$ as before. Note that these statistics are computed using the current (fixed) values of the parameters and follow from distribution (6) earlier.

In the M-step, $\langle \mathcal{L}_C \rangle$ is maximized with respect to $\mathbf{W}$ and σ^2 giving new parameter estimates

$$\tilde{\mathbf{W}} = \left\{ \sum_{n=1}^N (\mathbf{t}_n - \boldsymbol{\mu}) \langle \mathbf{x}_n \rangle^T \right\} \left(\sum_{n=1}^N \langle \mathbf{x}_n \mathbf{x}_n^T \rangle \right)^{-1}, \quad (27)$$

$$\tilde{\sigma}^2 = \frac{1}{Nd} \sum_{n=1}^N \{ \|\mathbf{t}_n - \boldsymbol{\mu}\|^2 - 2 \langle \mathbf{x}_n \rangle^T \tilde{\mathbf{W}}^T (\mathbf{t}_n - \boldsymbol{\mu}) + \text{tr}(\langle \mathbf{x}_n \mathbf{x}_n^T \rangle \tilde{\mathbf{W}}^T \tilde{\mathbf{W}}) \}. \quad (28)$$



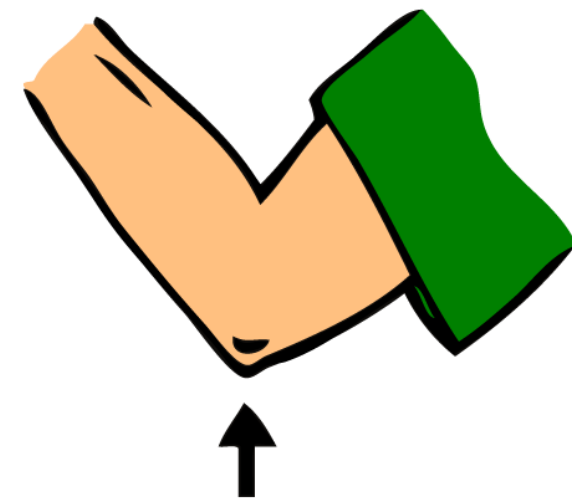
Expectation-Maximization (EM) algorithm

- The EM algorithm is an *iterative* method to find maximum likelihood estimates of model parameters, where the model often depends on unobserved latent variables Z .
- **Complete** likelihood:
$$\prod_{i=1}^n p(x_i | z_i) p(z_i)$$
- **Key idea 1**: Maximization of the **observed** likelihood $p(X; \theta)$ can be difficult but maximization of the **complete** likelihood $p(X, Z; \theta)$ can often be easier.
- **Observed** likelihood:
$$\prod_{i=1}^n \int p(x_i | z_i) p(z_i) dz_i$$

EM Algorithm

EM algorithm

$$\max_{\theta} \log p(x; \theta)$$



- Randomly initialize $\theta^{(0)}$.
- For iteration $t = 1, 2, \dots$
 - **Expectation (E) step:** derive the closed-form of the conditional expectation
$$Q(\theta | \theta^{(t)}) = \mathbb{E}_{Z|x, \theta^{(t)}} [\log p(x, Z; \theta)]$$
 - **Maximization (M) step:** maximize $Q(\theta | \theta^{(t)})$ to generate the next iterate
$$\theta^{(t+1)} = \arg \max_{\theta} Q(\theta | \theta^{(t)})$$

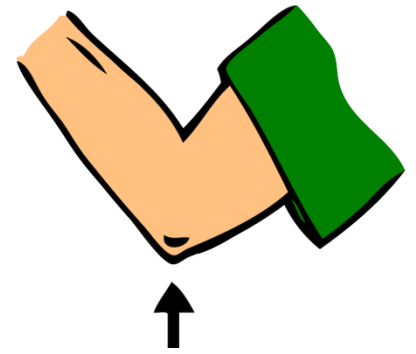
Key idea 2: “The ELBO”
(Evidence lower bound)

$$\begin{aligned} \log p(x; \theta) &\geq \underbrace{\mathbb{E}_{Z|x, \theta'} \left[\log \frac{p(x, Z; \theta)}{q(Z|x, \theta')} \right]}_{ELBO(x; \theta', \theta)} \\ &\geq \underbrace{\mathbb{E}_{Z|x, \theta'} [\log p(x, Z; \theta)]}_{Q(\theta|\theta')} \end{aligned}$$

Observed log
likelihood
Yucky.

Expected value
of the complete
log likelihood
Much nicer!

EM Algorithm



EM algorithm

$$\max_{\theta} \log p(x; \theta)$$

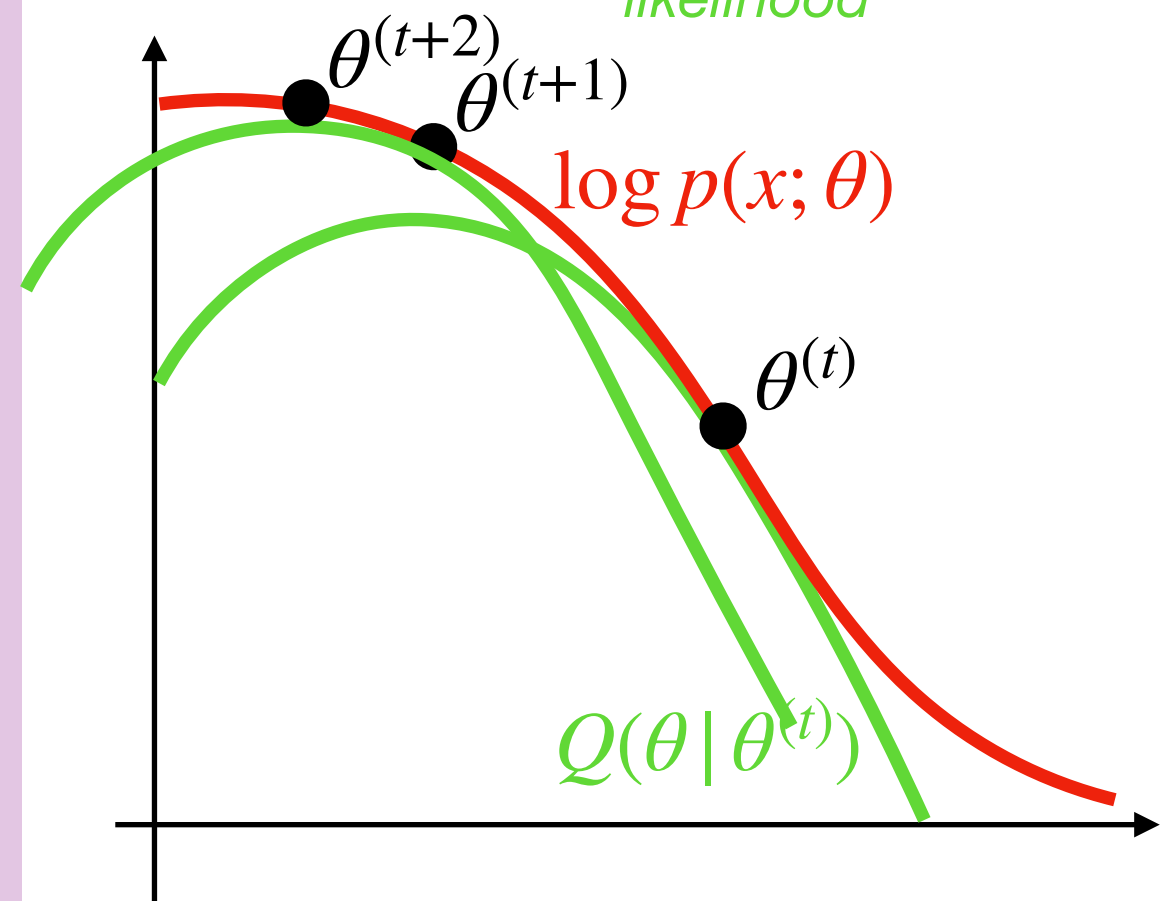
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Key idea 2: “The ELBO”
(Evidence lower bound)

$$\log p(x; \theta) \geq Q(\theta | \theta')$$

↑
Observed log
likelihood

↑
Expected value of the
complete log
likelihood



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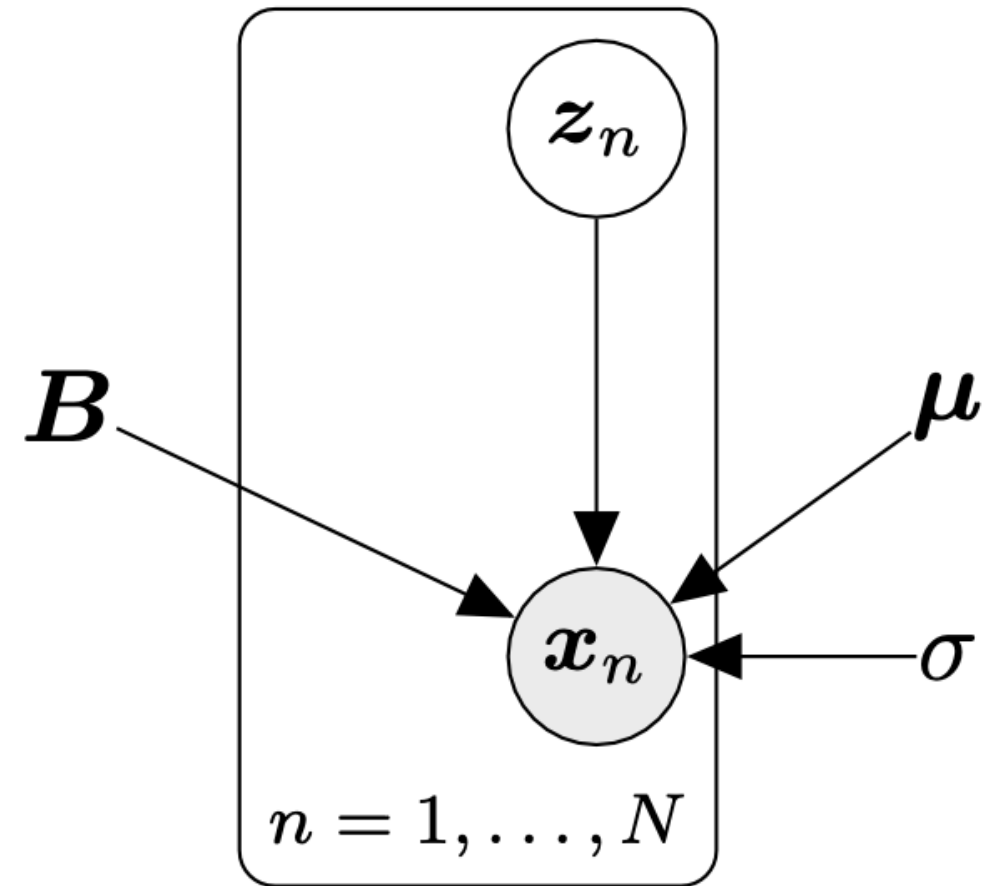
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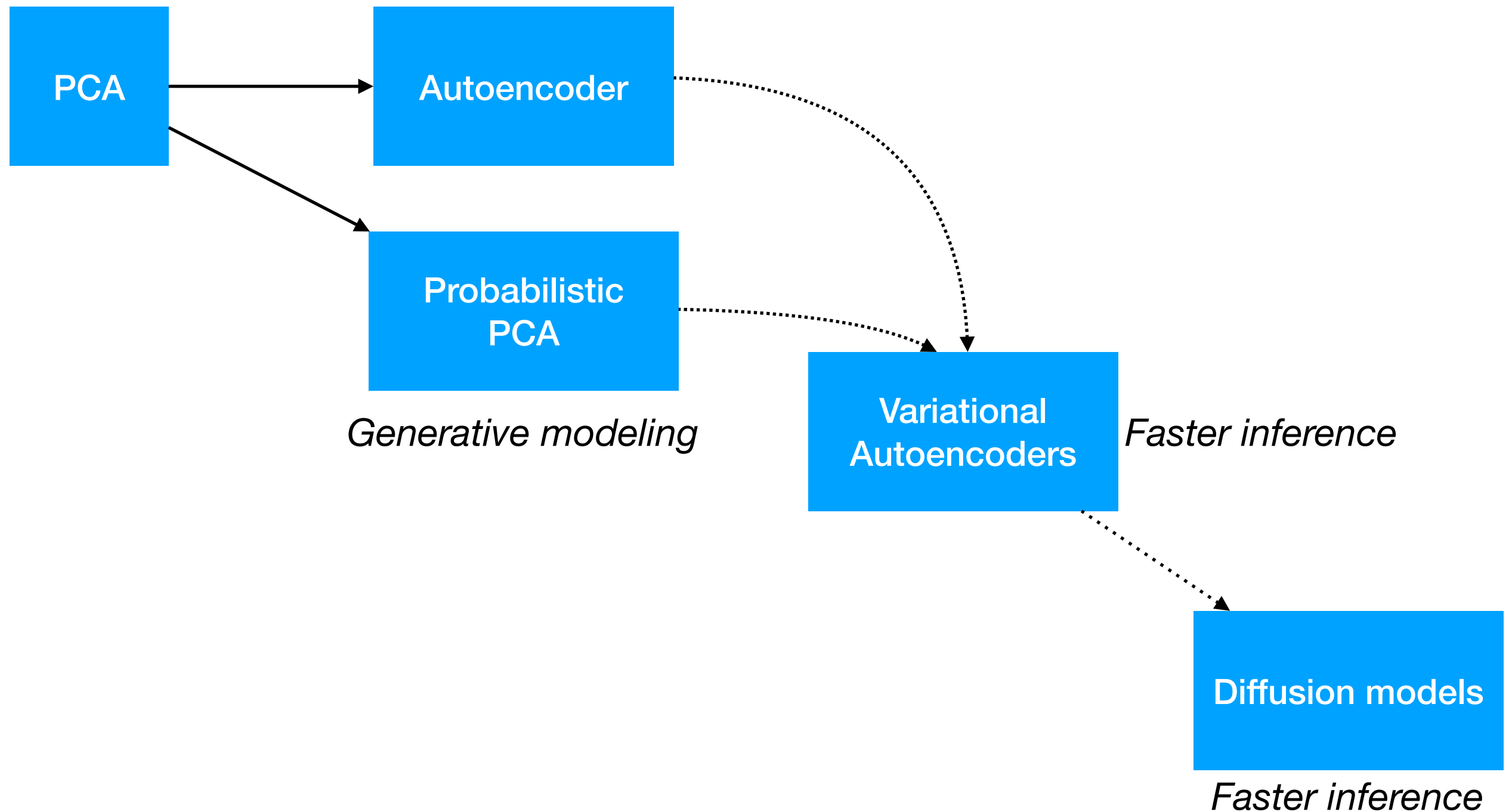
$$\tilde{\mathbf{W}} = \left\{ \sum_{n=1}^N (\mathbf{t}_n - \boldsymbol{\mu}) \langle \mathbf{x}_n \rangle^T \right\} \left(\sum_{n=1}^N \langle \mathbf{x}_n \mathbf{x}_n^T \rangle \right)^{-1}, \quad (27)$$

$$\tilde{\sigma}^2 = \frac{1}{Nd} \sum_{n=1}^N \{ \|\mathbf{t}_n - \boldsymbol{\mu}\|^2 - 2 \langle \mathbf{x}_n \rangle^T \tilde{\mathbf{W}}^T (\mathbf{t}_n - \boldsymbol{\mu}) + \text{tr}(\langle \mathbf{x}_n \mathbf{x}_n^T \rangle \tilde{\mathbf{W}}^T \tilde{\mathbf{W}}) \}. \quad (28)$$

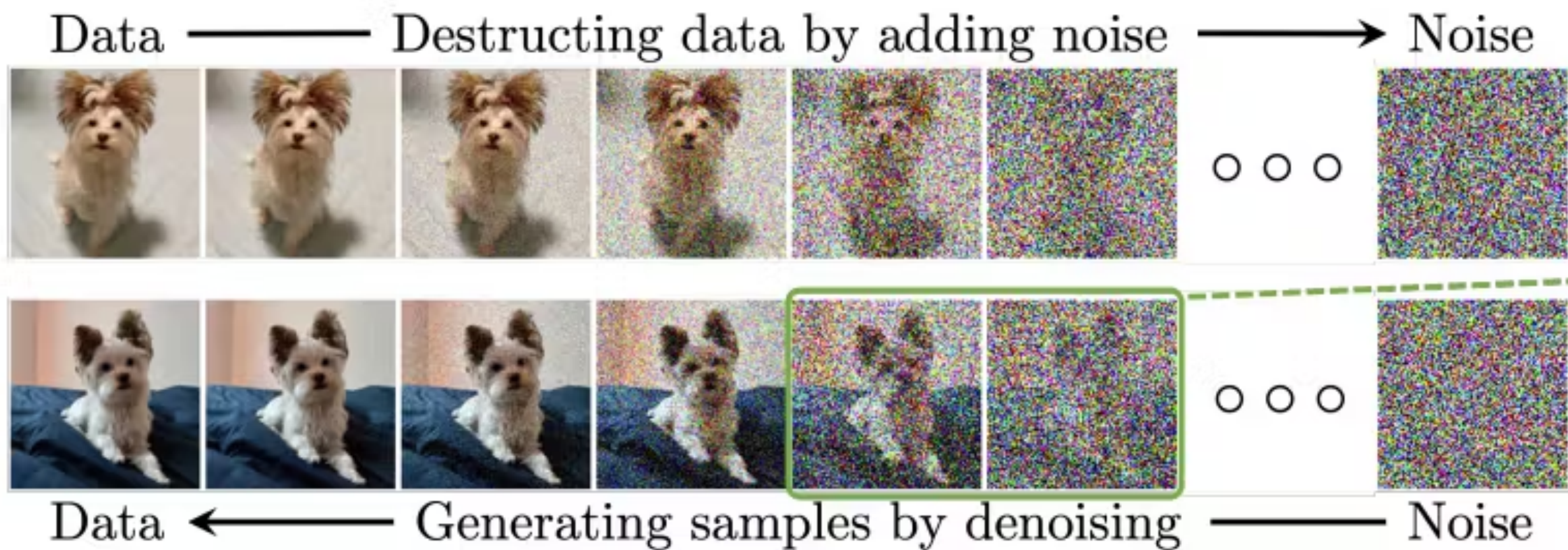


Outline of today's lecture

Nonlinear dimension reduction



Diffusion models



References

- **PCA:** ESL, ISL, MML
- **EM algorithm:**
 - <https://cs229.stanford.edu/notes2020spring/cs229-notes8.pdf>
- MML book Chapter 11.3